Bank of Korea G5 (Statistics) — Solutions

Practice problems for the written exam

Part I. Practice problems

A. Short-answer concepts

Solution 1 *Concepts I*

(a) The p-value is the probability, computed under H_0, of a test statistic at least as extreme as the one observed. Small values mean the data are unusual if H_0 holds; H_0 is rejected at level α iff $p \le \alpha$. It is a statement about the data given H_0, not about H_0 given the data: $P(H_0 \mid \text{data})$ needs a prior and Bayes's theorem, and can be large even when p is small (e.g. when H_1 is a priori implausible or the effect is tiny relative to n).

(b) T_n is unbiased if $ET_n = \theta$ for every n and θ; consistent if $T_n \xrightarrow{p} \theta$. $S_n^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ is biased ($ES_n^2 = \frac{n-1}{n}\sigma^2$) but consistent. Conversely $T_n = X_1$ is unbiased for μ but not consistent, since its distribution does not concentrate as n grows. A sufficient condition for consistency is bias $\to 0$ and variance $\to 0$ (MSE $\to 0$).

(c) A trend that holds within every subgroup can reverse when the groups are pooled, because the pooled comparison mixes groups in different proportions. Example: a treatment can have the higher success rate in both mild and severe cases yet the lower overall rate if it is given mostly to severe cases. The lesson is to condition on (adjust for) confounders.

(d) $\mathrm{Var}(\epsilon_i \mid x_i) = \sigma_i^2$ not constant. OLS stays unbiased and consistent but is no longer BLUE (GLS/WLS is), and the usual formula $\hat{\sigma}^2(X'X)^{-1}$ for its covariance is wrong, so t/F tests and confidence intervals are invalid. Remedies: WLS/GLS if the form is known, otherwise heteroskedasticity-consistent (White) standard errors; detection by residual plots, Breusch–Pagan or White tests.

(e) To assess the sampling distribution of a statistic $T(X_1, \ldots, X_n)$, draw B samples of size n *with replacement* from the observed data (i.e. from the empirical cdf $\hat{F}_n$), recompute T^{*b}, and use the spread of $T^{*1}, \ldots, T^{*B}$ to estimate the standard error, bias or confidence interval (e.g. percentile interval). It replaces analytic derivations by computation, relying on $\hat{F}_n \approx F$.

Solution 2 *Concepts II*

(a) $T(\mathbf{X})$ is sufficient for θ if the conditional distribution of $\mathbf{X}$ given T does not depend on θ: T carries all information about θ. Factorization theorem: T is sufficient iff $f(\mathbf{x};\theta) = g(T(\mathbf{x});\theta)h(\mathbf{x})$. E.g. for a Bernoulli sample $f = p^{\sum x_i}(1-p)^{n-\sum x_i}$, so $\sum X_i$ is sufficient.

(b) $\pi(\theta) = P_\theta(\text{reject } H_0)$. On H_0 it gives the type I error probabilities (its supremum there is the size); on H_1 it is the power $= 1 - P(\text{type II error})$. A good test has π small on Θ_0 and large on Θ_1; it increases with n and with the distance from H_0.

(c) Variables $x_t, y_t \sim I(1)$ are cointegrated if some linear combination $y_t - \beta x_t$ is $I(0)$: they share a stochastic trend and have a stable long-run relation. Then a regression of y on x is not spurious, and the dynamics have an error-correction representation $\Delta y_t = \alpha(y_{t-1} - \beta x_{t-1}) + \ldots$ (Granger representation theorem). Tests: Engle–Granger (unit-root test on OLS residuals with special critical values) and Johansen (rank of Π in a VECM).

(d) In $\Delta y_t = \mu(+\delta t) + \gamma y_{t-1} + \sum_{j=1}^{p} c_j \Delta y_{t-j} + \epsilon_t$, test $H_0 : \gamma = 0$ (unit root) against $\gamma < 0$ (stationary). The lagged differences absorb serial correlation. Under H_0 the t-ratio of $\hat{\gamma}$ does

not have a t distribution but the Dickey–Fuller distribution, whose critical values depend on the deterministic terms included. The test has low power against near-unit-root alternatives.

(e) A fixed factor has levels chosen deliberately and of interest in themselves; inference concerns their means $\mu + \alpha_i$ ($\sum \alpha_i = 0$). A random factor has levels sampled from a population of levels (days, batches, operators); $\alpha_i \sim N(0, \sigma_\alpha^2)$, and inference concerns the variance component σ_α^2 ($H_0 : \sigma_\alpha^2 = 0$). In one-way ANOVA the F statistic is the same, but the interpretation, the expected mean squares, and in multi-factor designs the correct denominators of F differ.

Solution 3 *Concepts III (official statistics)*

(a) With base-period quantities q_0: $P_L = \sum p_t q_0 / \sum p_0 q_0$; with current quantities: $P_P = \sum p_t q_t / \sum p_0 q_t$. Consumers substitute toward goods that became relatively cheaper, so the fixed base basket overweights items whose prices rose: Laspeyres tends to overstate and Paasche to understate the true cost-of-living change. Fisher's ideal index $\sqrt{P_L P_P}$ lies between. Laspeyres needs only base weights, which is why CPIs use it with periodic re-basing.

(b) A series is decomposed as trend-cycle × seasonal × irregular (or additively); seasonal adjustment removes the seasonal and calendar (trading-day, holiday) effects so that month-to-month movements are comparable. X-13 fits a regARIMA model to extend the series with forecasts and then applies symmetric moving-average filters (X-11) or model-based signal extraction (SEATS). Near the end of the series the filters rely on forecasts; as new data arrive the forecasts are replaced and the estimated seasonal factors change, so recent adjusted values are revised.

(c) deff $= \mathrm{Var}_{\mathrm{design}}(\hat\theta) / \mathrm{Var}_{\mathrm{SRS}}(\hat\theta)$ for the same sample size. Clustering gives deff > 1 (approximately $1 + (\bar m - 1)\rho$ with intraclass correlation ρ); stratification usually gives deff < 1; unequal weights increase it. The effective sample size is n/deff.

(d) AIC $= -2 \log \hat L + 2k$, BIC $= -2 \log \hat L + k \log n$ for a model with k parameters; choose the smallest. Both trade fit against complexity. BIC penalizes more for $n \geq 8$, selects smaller models, and is consistent (picks the true finite model with probability $\to 1$); AIC aims at predictive accuracy (estimates Kullback–Leibler risk) and may overfit asymptotically.

(e) Overfitting: a model flexible enough to fit noise has small training error but large error on new data (low bias, high variance). K-fold CV splits the data into K parts, fits on $K - 1$ parts, computes the prediction error on the held-out part, and averages over the K folds; the tuning parameter (model size, penalty) with the smallest CV error is chosen. $K = 5$ or 10 balances bias and variance of the error estimate.

B. Probability

Solution 4 *Bayes's theorem*

(a) $P(F) = 0.04(0.9) + 0.96(0.15) = 0.036 + 0.144 = 0.18$.

(b) $P(D \mid F) = 0.036/0.18 = 0.20$; $P(D \mid F^c) = 0.04(0.1)/0.82 = 0.0049$.

(c) The base rate is low, so even a 15% false-positive rate among the 96% healthy firms produces four times as many false flags (0.144) as true ones (0.036). With a 20% default rate, $P(F) = 0.18 + 0.12 = 0.30$ and $P(D \mid F) = 0.18/0.30 = 0.6$: the positive predictive value depends strongly on prevalence.

Solution 5 *Joint density*

(a) $f_X(x) = \int_0^1 (x+y)dy = x + \frac{1}{2}$. $EX = \int_0^1 x(x+\frac{1}{2})dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$, $EX^2 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$, $\operatorname{Var} X = \frac{5}{12} - \frac{49}{144} = \frac{11}{144}$.

(b) $E(XY) = \iint xy(x+y) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$, so Cov $= \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$ and Corr $= -\frac{1/144}{11/144} = -\frac{1}{11}$. Not independent ($f \neq f_X f_Y$; also Cov $\neq 0$).

(c) $P(X+Y<1) = \int_0^1 \int_0^{1-x} (x+y)dy\,dx = \int_0^1 \frac{1-x^2}{2}dx = \frac{1}{3}$. $E(Y \mid x) = \int_0^1 y\frac{x+y}{x+1/2}dy = \frac{x/2+1/3}{x+1/2} = \frac{3x+2}{6x+3}$.

Solution 6 *Limit theorems*

(a) $\operatorname{Var} N = 100$, so $P(|N-100| \geq 20) \leq 100/400 = 0.25$.

(b) $P(N \geq 120) = P(N > 119.5) \approx 1 - \Phi(19.5/10) = 1 - \Phi(1.95) = 0.0256$.

(c) $E\bar{X} = \mu$, $\operatorname{Var} \bar{X} = \sigma^2/n$; Chebyshev gives $P(|\bar{X} - \mu| \geq \epsilon) \leq \sigma^2/(n\epsilon^2) \to 0$. The WLLN is convergence in probability ($P(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ for each ϵ); the SLLN is almost-sure convergence ($P(\bar{X}_n \to \mu) = 1$), which is stronger, and needs only $E|X| < \infty$.

(d) X_n is a sum of n i.i.d. Poisson(1), so $(X_n - n)/\sqrt{n} \xrightarrow{d} N(0,1)$. Delta method with $g(x) = \sqrt{x}$, $g'(n) = 1/(2\sqrt{n})$: $\sqrt{X_n} - \sqrt{n} \approx (X_n - n)/(2\sqrt{n})$, so $2(\sqrt{X_n} - \sqrt{n}) \xrightarrow{d} N(0,1)$. The square root is variance-stabilizing: the limiting variance no longer depends on the mean.

Solution 7 *Order statistics*

(a) $P(X_{(1)} > t) = \prod P(X_i > t) = e^{-nt/\theta}$, so $X_{(1)} \sim$ Exp with mean θ/n; $E(nX_{(1)}) = \theta$ but $\operatorname{Var}(nX_{(1)}) = n^2(\theta/n)^2 = \theta^2$, compared with $\operatorname{Var} \bar{X} = \theta^2/n$: unbiased but not consistent.

(b) After the k-th failure, $n-k$ components remain, each with a fresh exponential life, so the next spacing is the minimum of $n-k$ exponentials, with mean $\theta/(n-k)$. Summing, $EX_{(n)} = \theta(\frac{1}{n} + \frac{1}{n-1} + \cdots + 1)$.

(c) $E = 2(1 + \frac{1}{2} + \frac{1}{3}) = \frac{11}{3} \approx 3.67$ years. $P(X_{(3)} > 4) = 1 - (1 - e^{-2})^3 = 1 - 0.8647^3 = 1 - 0.6465 = 0.354$.

C. Mathematical statistics

Solution 8 *Estimation for the uniform distribution*

(a) $EX = \theta/2$ gives $\tilde{\theta} = 2\bar{X}$. $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \geq x_{(n)}\}$ is decreasing on $[x_{(n)}, \infty)$, so $\hat{\theta} = X_{(n)}$.

(b) $\tilde{\theta}$ is unbiased with MSE $= 4\operatorname{Var} X/n = \theta^2/(3n)$. $X_{(n)}$ has density nt^{n-1}/θ^n, so $E\hat{\theta} = \frac{n}{n+1}\theta$, $E\hat{\theta}^2 = \frac{n}{n+2}\theta^2$, bias $-\theta/(n+1)$, and MSE $= \frac{2\theta^2}{(n+1)(n+2)}$. With $c = \frac{n+1}{n}$: $\operatorname{Var}(c\hat{\theta}) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{3n}$ for $n \geq 2$.

(c) Sufficiency: $f(\mathbf{x};\theta) = \theta^{-n}\mathbf{1}\{x_{(n)} \leq \theta\}$ (factorization). Completeness: if $\int_0^\theta g(t)nt^{n-1}\theta^{-n}dt = 0$ for all θ, differentiating gives $g(\theta)\theta^{n-1} = 0$, so $g = 0$ a.e. By Lehmann–Scheffé, $\frac{n+1}{n}X_{(n)}$ is the UMVUE.

(d) $P(n(\theta - \hat{\theta}) > x) = P(X_{(n)} < \theta - x/n) = (1 - \frac{x}{n\theta})^n \to e^{-x/\theta}$: the limit is exponential with mean θ. The support depends on θ, regularity conditions fail, and the MLE converges faster than the usual $\sqrt{n}$ rate.

Solution 9 *Neyman–Pearson and sample size*

(a) $\Lambda = \frac{L(104)}{L(100)} = \exp\{\frac{4}{100}\sum x_i - \text{const}\}$ is increasing in $\bar{x}$, so the MP test rejects for $\bar{x} > c$ with $P_{100}(\bar{X} > c) = 0.05$: $c = 100 + 1.645 \cdot 10/5 = 103.29$.

(b) Power $= P_{104}(\bar{X} > 103.29) = 1 - \Phi(\frac{103.29-104}{2}) = \Phi(0.355) = 0.639$. The rejection region does not depend on the particular $\mu_1 > 100$, so the test is UMP for $\mu > 100$ (the family has monotone likelihood ratio in $\bar{x}$). For $\mu \neq 100$ no UMP test exists: the MP region for $\mu_1 < 100$ is $\bar{x} < c'$.

(c) Need $\frac{4\sqrt{n}}{10} \geq z_{0.05} + z_{0.10} = 1.645 + 1.282$, so $n \geq (2.927 \cdot 2.5)^2 = 53.5$, i.e. $n = 54$.

Solution 10 *Likelihood ratio test for two exponential means*

(a) Unrestricted MLEs $\bar{x}, \bar{y}$; under H_0, $\hat{\theta}_0 = (n\bar{x} + m\bar{y})/(n+m)$. Then

$$\Lambda = \frac{\hat{\theta}_0^{-(n+m)} e^{-(n+m)}}{\bar{x}^{-n}\bar{y}^{-m}e^{-(n+m)}} = \frac{\bar{x}^n\bar{y}^m}{\hat{\theta}_0^{n+m}} = \frac{(n+m)^{n+m}T^n}{(nT+m)^{n+m}},$$

a function of T that is small when T is far from 1 in either direction.

(b) $2\sum X_i/\theta_1 \sim \chi^2_{2n}$ and $2\sum Y_j/\theta_2 \sim \chi^2_{2m}$, independent, so $\frac{\bar{X}/\theta_1}{\bar{Y}/\theta_2} \sim F(2n, 2m)$, which equals T under H_0.

(c) $T = 0.625$; reject if $T > 2.46$ or $T < 1/2.46 = 0.41$. Since $0.41 < 0.625 < 2.46$, do not reject (equal-tailed version of the LRT).

Solution 11 *Bayesian estimation*

(a) $\pi(\lambda \mid \mathbf{x}) \propto \lambda^{\sum x_i} e^{-n\lambda} \lambda^{\alpha-1} e^{-\beta\lambda}$, so $\lambda \mid \mathbf{x} \sim \text{Gamma}(\alpha + \sum x_i, \beta + n)$ and

$$E(\lambda \mid \mathbf{x}) = \frac{\alpha + \sum x_i}{\beta + n} = \frac{\beta}{\beta+n} \cdot \frac{\alpha}{\beta} + \frac{n}{\beta+n} \cdot \bar{x}.$$

(b) Posterior Gamma$(35, 12)$: mean $35/12 = 2.92$ (prior mean 2, MLE 3.1, weight on data $10/12$); 95% credible interval $(2.03, 3.96)$.

(c) The posterior median. A credible interval satisfies $P(\lambda \in C \mid \mathbf{x}) = 0.95$ for the observed data; a confidence interval is a random interval covering the fixed λ in 95% of repeated samples. With a vague prior they nearly coincide numerically.

Solution 12 *Rao–Blackwell and UMVUE*

(a) The Poisson family is a one-parameter exponential family with natural statistic T and parameter space containing an open interval, so T is complete and sufficient (directly: $Eg(T) = e^{-n\lambda}\sum g(t)(n\lambda)^t/t! = 0$ for all λ forces all coefficients $g(t) = 0$).

(b) Given $T = t$, $(X_1, \ldots, X_n)$ is multinomial$(t; \frac{1}{n}, \ldots, \frac{1}{n})$, so $X_1 \mid T = t \sim \text{Bin}(t, \frac{1}{n})$ and $E(\mathbf{1}\{X_1 = 0\} \mid T) = (1 - \frac{1}{n})^T$. By Lehmann–Scheffé this is the UMVUE.

(c) $(1-\frac{1}{n})^T = \exp\{T\log(1-\frac{1}{n})\} = \exp\{-\bar{X}(1 + \frac{1}{2n} + \ldots)\}$, so the two differ by $O_p(1/n)$ and have the same asymptotic distribution $N(e^{-\lambda}, \lambda e^{-2\lambda}/n)$.

Solution 13 *Fisher information and the delta method*

(a) $\ell = x\log p + (1-x)\log(1-p)$, $\ell'' = -x/p^2 - (1-x)/(1-p)^2$, $I(p) = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}$. CRLB $= p(1-p)/n = \operatorname{Var}\bar{X}$: attained.

(b) $g(p) = \log\frac{p}{1-p}$, $g'(p) = \frac{1}{p(1-p)}$. Delta method: $\sqrt{n}\{g(\bar{X}) - g(p)\} \xrightarrow{d} N(0, \frac{1}{p(1-p)})$.

(c) $\hat{\eta} = \log(0.3/0.7) = -0.847$, se $= 1/\sqrt{200 \cdot 0.21} = 0.154$; CI $-0.847 \pm 1.96(0.154) = (-1.150, -0.545)$. Back-transforming with $p = e^{\eta}/(1+e^{\eta})$: $(0.240, 0.367)$. Unlike the Wald interval for p, it always stays in $(0,1)$ and is asymmetric about $\hat{p}$.

D. Regression

Solution 14 *Simple linear regression from summary statistics*

(a) $\hat{\beta}_1 = 286/143 = 2$, $\hat{\beta}_0 = 20 - 2(7.5) = 5$. $SSR = \hat{\beta}_1 S_{xy} = 572$ (df 1), $SSE = 800 - 572 = 228$ (df 10), $MSE = 22.8$, $SST = 800$ (df 11); $R^2 = 0.715$.

(b) $\operatorname{se}(\hat{\beta}_1) = \sqrt{22.8/143} = 0.399$, $t = 5.01 > 2.228$; $F = 572/22.8 = 25.09 > 4.96$: reject. $t^2 = 25.09 = F$.

(c) β_1: $2 \pm 2.228(0.399) = (1.11, 2.89)$. At $x_0 = 15$, $\hat{y}_0 = 35$; $\operatorname{se}(\hat{y}_0) = \sqrt{22.8(\frac{1}{12} + \frac{56.25}{143})} = 3.30$, CI 35 ± 7.35; prediction se $= \sqrt{22.8 + 3.30^2} = 5.80$, PI 35 ± 12.93. A new observation carries its own error ϵ_0 with variance σ^2, which does not shrink with n; also $x_0 = 15$ is far from $\bar{x}$ (extrapolation), inflating both.

Solution 15 *The hat matrix and diagnostics*

(a) $H' = H$ is clear; $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$; $(I-H)^2 = I - 2H + H^2 = I - H$. $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = \operatorname{tr} I_p = p$.

(b) $(I-H)X = 0$, so $\mathbf{e} = (I-H)\epsilon$, $E\mathbf{e} = 0$, $\operatorname{Var}\mathbf{e} = (I-H)\sigma^2 I(I-H)' = \sigma^2(I-H)$. $E(\epsilon'(I-H)\epsilon) = \sigma^2\operatorname{tr}(I-H) = (n-p)\sigma^2$, so $s^2 = \mathbf{e}'\mathbf{e}/(n-p)$ is unbiased.

(c) $h_{ii} = x_i'(X'X)^{-1}x_i \in [0,1]$ measures how far x_i is from the centre of the design; $\sum h_{ii} = p$, and $h_{ii} > 2p/n$ is a common flag for high leverage. Studentized residual $r_i = e_i/(s\sqrt{1-h_{ii}})$ puts residuals on a common scale (outliers in y). Cook's distance $D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}$ measures the change in all fitted values when case i is deleted; it combines residual size and leverage to flag influential points ($D_i > 1$, or $> 4/n$).

Solution 16 *Multicollinearity and omitted variables*

(a) By the Frisch–Waugh theorem, $\hat{\beta}_j = \sum \tilde{x}_{ij}y_i / \sum \tilde{x}_{ij}^2$ where $\tilde{x}_j$ is the residual of x_j on the other regressors, and $\sum \tilde{x}_{ij}^2 = (1-R_j^2)\sum(x_{ij} - \bar{x}_j)^2$. Hence $\operatorname{Var}\hat{\beta}_j = \sigma^2/\{(1-R_j^2)S_{jj}\} = \frac{\sigma^2}{S_{jj}}\mathrm{VIF}_j$. With $R_j^2 = 0.9$, VIF $= 10$: the variance is ten times what it would be with orthogonal regressors (standard error $\times 3.16$). VIF > 10 is a common warning level.

(b) $\hat{\beta}_1 = \sum x_1 y / \sum x_1^2$ and $E\hat{\beta}_1 = \beta_1 + \beta_2 \sum x_1 x_2 / \sum x_1^2 = \beta_1 + \beta_2\delta$. No bias iff $\beta_2 = 0$ or $x_1 \perp x_2$ ($\delta = 0$).

(c) Drop or combine redundant variables (risk of omitted-variable bias); ridge regression (introduces bias, reduces variance; needs λ); principal component regression (coefficients harder to interpret; discarded components may matter for y); collect more or more varied data (costly).

Solution 17 *Heteroskedasticity and weighted least squares*

(a) $E\hat\beta_{\text{OLS}} = \beta$; $\text{Var} = \sum x_i^2\sigma^2 x_i^2/(\sum x_i^2)^2 = \sigma^2\sum x_i^4/(\sum x_i^2)^2$.

(b) Dividing by x_i: $y_i/x_i = \beta + u_i$ with $\text{Var}\,u_i = \sigma^2$, so the BLUE is $\hat\beta_W = \frac{1}{n}\sum y_i/x_i$, with variance σ^2/n. By Cauchy–Schwarz, $(\sum x_i^2)^2 \le n\sum x_i^4$, so $\sigma^2/n \le \text{Var}\,\hat\beta_{\text{OLS}}$.

(c) OLS $= 59.6/30 = 1.987$; WLS $= \frac{1}{4}(2.2 + 1.95 + 2.133 + 1.9) = 2.046$.

(d) $\hat\beta_{\text{GLS}} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y}$, with $\text{Var} = \sigma^2(X'V^{-1}X)^{-1}$; it is BLUE (Aitken's theorem): apply Gauss–Markov to the transformed model $V^{-1/2}\mathbf{y} = V^{-1/2}X\beta + V^{-1/2}\epsilon$, which has spherical errors.

Solution 18 *Logistic regression*

(a) $e^{0.47} = 1.60$: holding x_1 fixed, the odds of default are 60% higher for the self-employed. $e^{0.47} = 1.60$ also for a 10-point rise in debt-to-income: each 10 points multiplies the odds by 1.60.

(b) $G^2 = 120.4 - 101.2 = 19.2 > 5.99$ ($p < 0.001$): the predictors are jointly significant.

(c) $z = 0.47/0.30 = 1.57 < 1.96$: not significant at 5%.

(d) Fitted probabilities can fall outside $[0, 1]$; errors are heteroskedastic ($\text{Var} = p(1-p)$) and non-normal; the effect of x on p is forced to be constant, whereas it must flatten near 0 and 1.

Solution 19 *Structural change: Chow test*

(a) $F = \frac{(480-370)/3}{370/(45-6)} = \frac{36.67}{9.49} = 3.86 > 2.85$: reject equality; there is a structural break.

(b) With $D_t = 1$ after the crisis, fit $c_t = \beta_0 + \beta_1 y_t + \beta_2 r_t + \gamma_0 D_t + \gamma_1 D_t y_t + \gamma_2 D_t r_t + \epsilon_t$ and test $\gamma_0 = \gamma_1 = \gamma_2 = 0$ by the partial F test; the unrestricted SSE equals $SSE_1 + SSE_2$.

(c) Equal error variances in the two periods; check by $F = \frac{SSE_2/22}{SSE_1/17} = \frac{9.09}{10.0} = 0.91$ (Goldfeld–Quandt type). If the break point is unknown, use sup-F (Quandt–Andrews) tests.

E. Sampling

Solution 20 *Sample size for a proportion*

(a) $1.96\sqrt{p(1-p)/n} \le 0.03$ with the worst case $p = \frac{1}{2}$: $n_0 = 1.96^2(0.25)/0.03^2 = 1067.1$, so $n = 1068$.

(b) With the finite population correction, $n = n_0/\{1 + (n_0 - 1)/N\} = 1013.1$, so $n = 1014$.

(c) deff $= 1 + (20 - 1)(0.05) = 1.95$, so $n = 1068 \times 1.95 \approx 2083$ households (105 clusters).

Solution 21 *Ratio estimation*

(a) $\hat r = 52/40 = 1.3$, $\hat{\bar Y}_R = 1.3 \times 42 = 54.6$. $\hat V = \frac{1-f}{n}(s_y^2 + \hat r^2 s_x^2 - 2\hat r s_{xy}) = \frac{0.95}{25}(225 + 169 - 312) = 3.12$.

(b) $\hat V(\bar y) = \frac{0.95}{25}(225) = 8.55$: the ratio estimator is 2.7 times as efficient.

(c) $V(\bar y_R) - V(\bar y) \approx \frac{1-f}{n}(R^2 S_x^2 - 2RS_{xy}) < 0$ iff $\rho > \frac{RS_x}{2S_y} = \frac{1}{2}\frac{C_x}{C_y}$ (using $R = \bar Y/\bar X$). Here $\rho = 120/150 = 0.8 > \frac{1}{2}\cdot\frac{0.25}{0.288} = 0.43$.

Solution 22 *Stratified sampling and allocation*

Weights $W_h = (0.4, 0.3, 0.3)$, $f = 0.02$.

(a) $V_{\text{prop}} = \frac{1-f}{n}\sum W_h S_h^2 = \frac{0.98}{200}(40 + 120 + 480) = 3.14$.

(b) $n_h \propto N_h S_h = (40000, 60000, 120000)$: $n_h = (36.4, 54.5, 109.1) \approx (36, 55, 109)$. $V_{\text{Ney}} = \frac{(\sum W_h S_h)^2}{n} - \frac{\sum W_h S_h^2}{N} = \frac{22^2}{200} - \frac{640}{10000} = 2.42 - 0.064 = 2.36$.

(c) $S^2 \approx \sum W_h S_h^2 + \sum W_h(\bar{Y}_h - \bar{Y})^2 = 640 + 840 = 1480$ ($\bar{Y} = 80$), so $V_{\text{SRS}} = \frac{0.98}{200}(1480) = 7.25$. Proportional allocation removes the between-strata variation (gain $\times 2.3$); Neyman allocation also moves sample to the variable large-firm stratum (further gain $\times 1.33$).

Solution 23 *Cluster sampling*

(a) The cluster mean of M elements has variance $\frac{\sigma^2}{M}\{1 + (M-1)\rho\}$ because the $M(M-1)$ covariances each equal $\rho\sigma^2$. A sample of n clusters therefore has variance $\frac{\sigma^2}{nM}\{1 + (M-1)\rho\}$, against $\frac{\sigma^2}{nM}$ for an SRS of nM elements (ignoring fpc).

(b) Strata should be internally homogeneous and differ from each other, because only within-stratum variance enters the variance; clusters should be internally heterogeneous (each a miniature of the population), because the variance is driven by between-cluster differences. Clustering is used for cost and frame reasons (no list of elements needed); stratification for precision and domain estimates.

(c) deff $= 1 + 29(0.1) = 3.9$, effective size $600/3.9 \approx 154$.

F. Design of experiments

Solution 24 *One-way ANOVA with unequal replication*

(a) $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$, $\sum n_i \alpha_i = 0$, $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Means $20.5, 26, 20.67$, grand mean 22.83. $SS_A = 86.0$ (df 2), $SS_E = 27.67$ (df 9), $SS_T = 113.67$ (df 11); $MS_A = 43.0$, $MS_E = 3.07$, $F = 13.99 > 4.26$: reject.

(b) $5.5 \pm 2.262\sqrt{3.07(\frac{1}{5} + \frac{1}{4})} = 5.5 \pm 2.66 = (2.84, 8.16)$. With several intervals each at 95%, the probability that all cover is below 95% (familywise error); Tukey's HSD or Bonferroni keeps the simultaneous level.

(c) Residual plots against fitted values and groups (equal variance; Levene/Bartlett tests), normal Q–Q plot of residuals, and the randomization/run order for independence.

Solution 25 *Randomized complete block design*

(a) $y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$, $\sum \tau_i = \sum \beta_j = 0$, $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Treatment means $14.5, 17, 12.25$; block means $12, 15, 14.67, 16.67$; grand mean 14.583. $SS_{Tr} = 45.17$ (df 2), $SS_{Bl} = 33.58$ (df 3), $SS_E = 2.17$ (df 6), $SS_T = 80.92$ (df 11). $F_{Tr} = \frac{22.58}{0.361} = 62.5 > 5.14$: the fertilizers differ.

(b) CRD error: $(33.58 + 2.17)/9 = 3.97$ against 0.361. $RE = \frac{(b-1)MS_{Bl} + b(a-1)MS_E}{(ab-1)MS_E} = \frac{33.58 + 2.89}{3.97} = 9.2$: the CRD would need about nine times as many plots for the same precision.

(c) With one observation per cell, the interaction and error have the same degrees of freedom and cannot be separated; additivity must be assumed (Tukey's one-degree-of-freedom test can check one form of non-additivity).

Solution 26 *2^2 factorial design*

(a) $A = \frac{a+ab-b-(1)}{2r} = \frac{26}{4} = 6.5$, $B = \frac{b+ab-a-(1)}{2r} = \frac{14}{4} = 3.5$, $AB = \frac{ab+(1)-a-b}{2r} = \frac{6}{4} = 1.5$. $SS = \frac{(\text{contrast})^2}{4r}$: $SS_A = 84.5$, $SS_B = 24.5$, $SS_{AB} = 4.5$.

(b) $MS_E = 1.5$; $F_A = 56.3$, $F_B = 16.3$ significant; $F_{AB} = 3.0 < 7.71$ not significant.

(c) Interaction means the effect of A depends on the level of B (here A's effect is 5 at low B and 8 at high B in mean units). When interaction is large, an averaged main effect may describe no actual condition, and one must compare cell means (simple effects).

Solution 27 *Random-effects model*

(a) $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$, $\alpha_i \overset{\text{iid}}{\sim} N(0, \sigma_\alpha^2)$ independent of $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. $E(MS_E) = \sigma^2$, $E(MS_A) = \sigma^2 + b\sigma_\alpha^2$ with $b = 4$.

(b) $F = 60/12 = 5 > 3.06$: batch variation is significant. $\hat{\sigma}^2 = 12$, $\hat{\sigma}_\alpha^2 = (60-12)/4 = 12$, ICC $= 12/24 = 0.5$.

(c) $\text{Var}(\bar{y}_{\cdot\cdot}) = \frac{\sigma_\alpha^2}{5} + \frac{\sigma^2}{20} = \frac{E(MS_A)}{20}$, estimated by $60/20 = 3$. More batches reduce it more than more units per batch.

G. Time series

Solution 28 *AR(1): moments and forecasting*

(a) $EX_t = 2$, $\gamma_0 = \sigma^2/(1-\phi^2) = 1.5625$, $\rho_k = 0.6^k$ (geometric decay). PACF $\phi_{11} = 0.6$ and $\phi_{kk} = 0$ for $k \geq 2$.

(b) $\hat{X}_T(h) = \mu + \phi^h(X_T - \mu)$: 3.8 and 3.08. Error variances $\sigma^2 \sum_{j<h} \phi^{2j}$: 1 and 1.36. Intervals 3.8 ± 1.96 and 3.08 ± 2.29. As $h \to \infty$ the forecast tends to $\mu = 2$ and the variance to $\gamma_0 = 1.5625$.

(c) Multiply by $X_{t-1} - \mu$ and take expectations: $\gamma_1 = \phi\gamma_0$, so $\hat{\phi} = \hat{\rho}_1 = \sum(x_t - \bar{x})(x_{t-1} - \bar{x}) / \sum(x_t - \bar{x})^2$; then $\hat{\sigma}^2 = \hat{\gamma}_0(1 - \hat{\phi}^2)$.

Solution 29 *MA(1) and ARMA(1,1)*

(a) $\gamma_0 = (1+\theta^2)\sigma^2$, $\gamma_1 = -\theta\sigma^2$, $\gamma_k = 0$ ($k \geq 2$): $\rho_1 = -\theta/(1+\theta^2)$, unchanged by $\theta \to 1/\theta$. Only for $|\theta| < 1$ can ϵ_t be written as a convergent combination of current and past X's ($\epsilon_t = \sum \theta^j X_{t-j}$), which makes the model identifiable and forecasting possible. $\theta = 0.5$: $\rho_1 = -0.4$.

(b) $1 + \theta^2 \geq 2|\theta|$.

(c) $(1 - \phi B)X_t = (1 - \theta B)\epsilon_t$ gives $\psi_0 = 1$, $\psi_j = (\phi - \theta)\phi^{j-1}$ for $j \geq 1$.

(d) AR(p): ACF decays, PACF cuts off after lag p. MA(q): ACF cuts off after q, PACF decays. ARMA: both decay. In practice the patterns are blurred by sampling error, so candidate models are compared by AIC/BIC and residual diagnostics (Ljung–Box).

Solution 30 *Unit roots and spurious regression*

(a) $y_t = \delta t + \sum_{s \leq t} \epsilon_s$: $Ey_t = \delta t$, $\text{Var}\, y_t = t\sigma^2$, both depending on t. Differencing gives $\Delta y_t = \delta + \epsilon_t$, which is stationary: $y_t \sim I(1)$.

(b) The t statistic diverges, so the slope looks "significant" far more often than the nominal 5%; R^2 does not tend to 0; residuals are $I(1)$ and DW $\to 0$. A high R^2 with very low DW is the classic warning sign (Granger–Newbold). Remedies: difference the data, or test for cointegration and use an ECM.

(c) Trend-stationary: $y_t = a + bt + u_t$ with stationary u_t; shocks are transitory and detrending is appropriate. Difference-stationary: shocks are permanent. Differencing a trend-stationary series creates a non-invertible MA unit root; detrending a difference-stationary series leaves a nonstationary residual and spurious cycles.

Solution 31 *VAR and Granger causality*

(a) The eigenvalues of A solve $\lambda^2 - 0.9\lambda + 0.14 = 0$: $\lambda = 0.7, 0.2$, both inside the unit circle, so stationary.

(b) The shock $e_2 = (0,1)'$ gives $Ae_2 = (0.2, 0.4)'$ and $A^2 e_2 = (0.18, 0.22)'$, since $A^2 = \begin{pmatrix} 0.31 & 0.18 \\ 0.27 & 0.22 \end{pmatrix}$.

(c) x Granger-causes y if past x improves the prediction of y given past y. Here: $H_0 : A_{12} = 0$, tested by a Wald/F test in the inflation equation.

(d) Reduced-form errors are correlated, so a "shock to one variable alone" is not meaningful; Cholesky orthogonalization imposes a recursive contemporaneous ordering. Results can depend on that ordering, which needs economic justification.

Solution 32 *Exponential smoothing and seasonality*

(a) For the model, $\hat{y}_{t+1} = y_t - \theta\epsilon_t$ and $\epsilon_t = y_t - \hat{y}_t$, so $\hat{y}_{t+1} = y_t - \theta(y_t - \hat{y}_t) = (1-\theta)y_t + \theta\hat{y}_t$.

(b) $(1-B)(1-B^{12})y_t = (1-\theta B)(1-\Theta B^{12})\epsilon_t$: regular and seasonal differencing remove trend and stable seasonality; the two MA factors model short-run and year-to-year correlation of shocks. It is the default regARIMA model in X-13.

(c) $\Delta_{12} \log y_t$ is the year-on-year growth rate; both remove stable seasonality, at the cost of a lag (they reflect developments averaged over 12 months), which is why seasonally adjusted month-on-month rates are also published.

Solution 33 *GARCH and volatility*

(a) $Er_t^2 = \omega/(1-\alpha-\beta) = 0.02/0.02 = 1$. Persistence $\alpha + \beta = 0.98$: a shock to volatility decays like 0.98^h (half-life $\log 0.5 / \log 0.98 \approx 34$ days).

(b) $E(r_t \mid \mathcal{F}_{t-1}) = 0$ gives $E(r_t r_{t-k}) = 0$; but r_t^2 depends on r_{t-1}^2, so $\mathrm{Cov}(r_t^2, r_{t-1}^2) \neq 0$ (volatility clustering).

(c) $Er_t^4 = 3E\sigma_t^4$ and $E\sigma_t^4 \geq (E\sigma_t^2)^2$, with strict inequality since σ_t^2 is random. Explicitly, with $m_2 = \omega/(1-\alpha)$, $Er^4 = 3(\omega^2 + 2\omega\alpha m_2)/(1-3\alpha^2)$, and kurtosis $= 3\frac{1-\alpha^2}{1-3\alpha^2} > 3$.

H. Multivariate analysis and computing

Solution 34 *Principal components*

(a) The 2×2 block has eigenvalues $6, 2$ with eigenvectors $(1,1)/\sqrt{2}$, $(1,-1)/\sqrt{2}$; the third is 1 with e_3. $Y_1 = (X_1+X_2)/\sqrt{2}$ (variance 6, 66.7%), $Y_2 = (X_1-X_2)/\sqrt{2}$ (2, 22.2%), $Y_3 = X_3$ (1, 11.1%).

(b) $R = \begin{pmatrix} 1 & 0.5 & 0 \\ 0.5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$: eigenvalues $1.5, 1, 0.5$, components $(X_1^* + X_2^*)/\sqrt{2}$ (50%), X_3^* (33.3%), $(X_1^* - X_2^*)/\sqrt{2}$ (16.7%). Standardizing removes the effect of scale: X_3 was "unimportant" in (a) only because of its small variance. Use R when units differ.

(c) Cumulative proportion of variance (e.g. $\geq 80\%$), Kaiser's rule (eigenvalues of R above 1), the scree plot elbow, or cross-validated prediction error when PCs feed a regression.

Solution 35 *Linear discriminant analysis*

(a) Assign to group 2 iff $f_2(\mathbf{x}) > f_1(\mathbf{x})$; the quadratic terms cancel, leaving $a'\{\mathbf{x} - \frac{1}{2}(\mu_1+\mu_2)\} > 0$ with $a = \Sigma^{-1}(\mu_2 - \mu_1)$.

(b) $a = \Sigma^{-1}(2,1)' = (2,0)'$, so the rule is $2x_1 - 2 > 0$, i.e. $x_1 > 1$. $\Delta^2 = (2,1)a = 4$, $\Delta = 2$; error probability $\Phi(-\Delta/2) = \Phi(-1) = 0.159$ for each group.

(c) With priors: threshold $\log(\pi_1/\pi_2)$ on the right-hand side. With unequal Σ_k: quadratic discriminant analysis, since the quadratic terms no longer cancel.

Solution 36 *Monte Carlo integration and Newton–Raphson*

(a) $\operatorname{Var} e^U = \frac{e^2-1}{2} - (e-1)^2 = 0.2420$, so $\operatorname{Var}\hat{\theta} = 0.242/n$; $n = 0.242/10^{-6} = 242{,}000$.

(b) $\operatorname{Cov}(e^U, e^{1-U}) = e - (e-1)^2 = -0.2342$. For each pair, $\operatorname{Var}(e^U + e^{1-U}) = 2(0.242) - 2(0.2342) = 0.0157$, so with n evaluations the variance is $\frac{0.0157}{2n} = \frac{0.0078}{n}$, about 31 times smaller. The integrand is monotone, so e^U and e^{1-U} are negatively correlated and their errors partly cancel.

(c) $\theta^{(k+1)} = \theta^{(k)} - \ell'(\theta^{(k)})/\ell''(\theta^{(k)})$, iterated until $|\ell'|$ is small; it converges quadratically near the maximum but may fail from poor starts. Fisher scoring replaces $-\ell''$ by its expectation $nI(\theta)$, which is always positive and often simpler (for exponential families with canonical link the two coincide; for GLMs it is IRLS).

Solution 37 *Statistical learning*

(a) $E\{(Y - \hat{f}(x_0))^2\} = \sigma^2 + \{E\hat{f}(x_0) - f(x_0)\}^2 + \operatorname{Var}\hat{f}(x_0)$: irreducible error, squared bias, variance. Flexible models lower bias but raise variance.

(b) Ridge: $\min\|\mathbf{y} - X\beta\|^2 + \lambda\sum\beta_j^2$; lasso: $\min\|\mathbf{y} - X\beta\|^2 + \lambda\sum|\beta_j|$. The ℓ_1 constraint region has corners on the axes, so the solution often lies at a corner with some $\hat{\beta}_j = 0$ exactly; the ℓ_2 ball is smooth and only shrinks coefficients. (In the orthonormal case lasso soft-thresholds, ridge rescales by $1/(1+\lambda)$.)

(c) Accuracy $860/1000 = 0.86$, precision $80/200 = 0.40$, recall $80/100 = 0.80$, specificity $780/900 = 0.867$. With 10% positives, the trivial rule "always negative" already has accuracy 0.90, so precision/recall (or AUC) are more informative.

I. Essay topics (Part II type)

Solution 38 *Essay: alternative data and nowcasting*

Outline. (1) Benefits: timeliness (daily/weekly), granularity (regions, industries), coverage of new activities; useful for turning points (e.g. the 2020 pandemic shock). (2) Statistical issues: selection bias (card users and platforms are not the population), unstable coverage and definitions over time (structural breaks when providers change), noise and seasonality (calendar effects need adjustment), spurious correlation and overfitting with many predictors (Google Flu Trends), privacy and data access continuity. (3) Methods: bridge equations and MIDAS for mixed frequencies; dynamic factor models with Kalman filter to handle ragged edges; shrinkage (lasso, ridge) and out-of-sample evaluation; benchmarking/calibration to official totals; publishing uncertainty bands. (4) Position: use as a complement to, not substitute for, official statistics; document methods and revisions; evaluate forecasts in real time.

Solution 39 *Essay: revisions of official statistics*

Outline. (1) Sources: incomplete source data in early estimates (extrapolation, imputation), late survey responses, benchmark revisions (new census/input–output tables, base-year changes, chain-linking), concurrent seasonal adjustment (end-point problem of filters), methodological and classification changes. (2) Trade-off: early estimates are valuable for policy but less precise; measure accuracy by revision statistics (mean revision for bias, mean absolute revision for size), and compare with the policy cost of delay. (3) Practice: a published revision calendar and policy, real-time databases (vintages) for users, explicit uncertainty measures (fan charts), seasonal-adjustment options that reduce revisions (forecast extension, controlled updating of models), clear explanation of major revisions. (4) Conclusion: revisions are a normal consequence of producing timely statistics; transparency, not their absence, sustains trust.

Solution 40 *Essay: survey nonresponse*

Outline. (1) Bias of the respondent mean: $\text{bias}(\bar{y}_R) \approx \frac{\text{Cov}(\rho,y)}{\bar{\rho}}$, where ρ is the response propensity: low response rates do not cause bias by themselves; correlation between propensity and the variable does. MCAR/MAR/MNAR framework. (2) Design-stage: callbacks, mixed modes (web, phone, face-to-face), incentives, responsive designs that target low-propensity groups, panel maintenance, shorter questionnaires. (3) Estimation-stage: weighting-class adjustment and propensity-score weighting, calibration (raking, GREG) to known totals (census, administrative data), imputation (hot-deck, regression, multiple imputation for item nonresponse), sensitivity analysis for MNAR. (4) Report response rates and R-indicators, and evaluate with administrative linkage where possible.

Part II. Past exam problems (restated)

2026 round (exam of 13 September 2025)

Past problem 1 *[2026 #1]*

(a) Rejecting H_0 when it is true; its maximal probability over H_0 is the size α, controlled by the choice of the rejection region. (b) A random interval $(L(\mathbf{X}), U(\mathbf{X}))$ with $P_\theta(L \le \theta \le U) = 1-\alpha$ for all θ; in repeated sampling $100(1-\alpha)\%$ of such intervals cover θ (the realized interval either covers or not). (c) Regular, calendar-related fluctuation within a year (months/quarters) that recurs with roughly stable timing and size, caused by climate, holidays, institutional schedules; removed

by seasonal adjustment or modelled by seasonal dummies/SARIMA. (d) Write an integral as an expectation $\theta = Eh(X)$ and estimate it by $\frac{1}{n}\sum h(X_i)$ from simulated draws; the error is $O_p(n^{-1/2})$ regardless of dimension, and variance can be reduced by importance sampling, antithetic or control variates. (e) Find orthogonal linear combinations $Y_k = \mathbf{v}_k'\mathbf{X}$ with maximal variance successively, given by the eigenvectors of the covariance (or correlation) matrix; the eigenvalues are the variances. Used for dimension reduction and to handle multicollinearity.

Past problem 2 *[2026 #2]*

(a) $\hat{V}(\bar{y}_3) = (1 - \frac{8}{100})\frac{85.6}{8} = 9.84$, se 3.14: $55.6 \pm 6.27 = (49.32, 61.88)$.

(b) $W_h = (0.5, 0.25, 0.25)$: $\bar{y}_{st} = 8.65 + 8.3 + 13.9 = 30.85$. $\hat{V} = \sum W_h^2(1 - f_h)\frac{s_h^2}{n_h} = 0.25(3.492) + 0.0625(2.88) + 0.0625(9.844) = 1.67$ (se 1.29).

(c) Minimizing V subject to cost gives $n_h \propto N_h\sigma_h/\sqrt{c_h} = (250, 250, 200)$ and $n = \dfrac{C\sum N_h\sigma_h/\sqrt{c_h}}{\sum N_h\sigma_h\sqrt{c_h}} = \dfrac{900 \cdot 700}{10000} = 63$. So $n_h = (22.5, 22.5, 18)$; rounding within budget, e.g. $(22, 23, 18)$ costing 894.

Past problem 3 *[2026 #3]*

(a) $R^2 = SSR/(SSR + SSE)$. Under H_0, $SSR/\sigma^2 \sim \chi^2_p$ and $SSE/\sigma^2 \sim \chi^2_{n-p-1}$ independently (Cochran's theorem: they are quadratic forms in orthogonal projections $H - \frac{1}{n}\mathbf{1}\mathbf{1}'$ and $I - H$ applied to $\mathbf{y}$, and $\mathbf{y}$ has mean in the span of $\mathbf{1}$). If $U \sim \chi^2_{2a}$, $V \sim \chi^2_{2b}$ independent, $U/(U+V) \sim \text{Beta}(a, b)$ (Gamma–Beta relation), so $R^2 \sim \text{Beta}(p/2, (n-p-1)/2)$.

(b) Writing $W = U/(U+V)$, $\frac{W/a}{(1-W)/b} = \frac{U/2a}{V/2b} \sim F(2a, 2b)$. With $a = p/2$, $b = (n-p-1)/2$: $F = \frac{R^2/p}{(1-R^2)/(n-p-1)} = \frac{SSR/p}{SSE/(n-p-1)} \sim F(p, n-p-1)$; reject H_0 if $F > F_\alpha(p, n-p-1)$.

(c) $nR^2 = \frac{SSR}{\sigma^2} \cdot \frac{n\sigma^2}{SST}$. Under H_0, $SST/(n-1) = s_y^2 \xrightarrow{p} \sigma^2$, so the second factor $\xrightarrow{p} 1$, and $SSR/\sigma^2 \sim \chi^2_p$; by Slutsky $nR^2 \xrightarrow{d} \chi^2_p$ (the basis of LM tests).

Past problem 4 *[2026 #4]*

(a) $E\exp\{s\bar{X} + \sum t_i(X_i - \bar{X})\} = E\exp\{\sum c_i X_i\}$ with $c_i = \frac{s}{n} + t_i - \bar{t}$. Since $\sum c_i = s$ and $\sum c_i^2 = \frac{s^2}{n} + \sum(t_i - \bar{t})^2$, it equals $\exp\{\mu s + \frac{\sigma^2 s^2}{2n}\} \cdot \exp\{\frac{\sigma^2}{2}\sum(t_i - \bar{t})^2\}$, the product of the mgf of $\bar{X}$ and a function of $\mathbf{t}$ alone (the mgf of T): independence.

(b) $f(\mathbf{y};\theta) = (2\pi)^{-n/2}\exp\{-\frac{1}{2}\sum y_i^2\}\exp\{n\theta\bar{y} - \frac{n}{2}\theta^2\}$: a full-rank one-parameter exponential family with natural statistic $\bar{Y}$, hence complete and sufficient.

(c) By Rao–Blackwell and Lehmann–Scheffé, the UMVUE is $E\{\mathbf{1}(Y_1 > a) \mid \bar{Y}\}$. $Y_1 - \bar{Y} \sim N(0, \frac{n-1}{n})$ is independent of $\bar{Y}$ by (a), so

$$\hat{\eta} = P(Y_1 - \bar{Y} > a - \bar{y} \mid \bar{Y} = \bar{y}) = 1 - \Phi\Big(\frac{a - \bar{Y}}{\sqrt{(n-1)/n}}\Big) = \Phi\Big(\sqrt{\tfrac{n}{n-1}}(\bar{Y} - a)\Big).$$

Past problem 5 *[2026 #5]*

(a) $y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$, $\sum \alpha_i = \sum \beta_j = 0$, $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, no interaction. $S_A = 16.67$ (3), $S_B = 12.5$ (2), $S_E = 6.83$ (6), $S_T = 36$ (11); $V_A = 5.56$, $V_B = 6.25$, $V_E = 1.139$; $F_A = 4.88$, $F_B = 5.49$.

(b) $F_{0.05}(3,6) = 4.76$, $F_{0.05}(2,6) = 5.14$: both significant (barely).

(c) $S_E(y) = \sum\sum y_{ij}^2 - \frac{1}{b}\sum T_{i\cdot}^2 - \frac{1}{a}\sum T_{\cdot j}^2 + \frac{T^2}{ab}$ is quadratic in y. Setting $dS_E/dy = 0$ gives Yates's formula $y = \dfrac{aT'_{i\cdot} + bT'_{\cdot j} - T'}{(a-1)(b-1)}$, with totals excluding the missing cell: $T'_{3\cdot} = 11$, $T'_{\cdot 2} = 11$, $T' = 43$, so $y = \frac{4(11)+3(11)-43}{6} = 5.67$.

(d) $S_E = 37.75 - 18.56 - 12.57 = 6.62$ with $6 - 1 = 5$ df (one df lost for the estimated value), $V_E = 1.324$. $F_A = 6.19/1.324 = 4.67 < F_{0.05}(3,5) = 5.41$, $F_B = 6.285/1.324 = 4.75 < F_{0.05}(2,5) = 5.79$: neither is significant now. Losing a degree of freedom raises the critical values, so the borderline effects of (b) are no longer significant (total df 10).

Past problem 6 *[2026 #6]*

(a) The map $(\alpha, \beta) \mapsto (\mu, \sigma^2) = (\alpha, \alpha\beta)$ is one-to-one, and the normal MLE is $(\bar{x}, \hat{\sigma}^2)$; by invariance $\hat{\alpha} = \bar{x}$, $\hat{\beta} = \hat{\sigma}^2/\bar{x}$.

(b) With $\beta = 1$, $\ell(\alpha) = -\frac{n}{2}\log\alpha - \frac{\sum(x_i-\alpha)^2}{2\alpha}$ ($\alpha > 0$). $\ell'(\alpha) = 0$ gives, after multiplying by $2\alpha^2$, $-n\alpha + \sum x_i^2 - n\alpha^2 = 0$, i.e. $\alpha^2 + \alpha - m_2 = 0$ with $m_2 = \frac{1}{n}\sum x_i^2$: $\hat{\alpha}_R = \frac{-1+\sqrt{1+4m_2}}{2}$.

(c) $\sup_{H_0} L = (2\pi\hat{\alpha}_R)^{-n/2}\exp\{-\sum(x_i - \hat{\alpha}_R)^2/(2\hat{\alpha}_R)\}$ and $\sup L = (2\pi\hat{\sigma}^2)^{-n/2}e^{-n/2}$; their ratio is the stated $\lambda(\mathbf{x})$. Reject for small λ; asymptotically $-2\log\lambda \xrightarrow{d} \chi_1^2$.

Past problem 7 *[2026 #7]*

(a) $(1 - \beta_1 B)\sigma_t^2 = \alpha_0 + \alpha_1\varepsilon_{t-1}^2$, so $\sigma_t^2 = \frac{\alpha_0}{1-\beta_1} + \alpha_1\sum_{k\geq 1}\beta_1^{k-1}\varepsilon_{t-k}^2$: ARCH($\infty$) with geometrically declining weights $\gamma_k = \alpha_1\beta_1^{k-1}$. Advantage: long memory in volatility with only three parameters (parsimony, easier to satisfy positivity and stationarity constraints).

(b) Taking expectations, $E\varepsilon_t^2 = E\sigma_t^2 = \alpha_0 + (\alpha_1 + \beta_1)E\varepsilon_{t-1}^2$, so $\operatorname{Var}\varepsilon_t = \alpha_0/(1 - \alpha_1 - \beta_1)$.

(c) Substitute $\sigma_t^2 = \varepsilon_t^2 - \nu_t$: $\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1)\varepsilon_{t-1}^2 + \nu_t - \beta_1\nu_{t-1}$. ν_t is a martingale difference (white noise), so ε_t^2 is ARMA(1,1) with AR coefficient $\alpha_1 + \beta_1$ and MA coefficient $-\beta_1$.

Past problem 8 *[2026 #8]*

(a) $\ell = \sum\{y_i\eta_i - \log(1 + e^{\eta_i})\}$ with $\eta_i = \beta_0 + \beta_1 x_i$, and $\frac{d}{d\eta}\log(1 + e^{\eta}) = p$; the chain rule gives both.

(b) $\ell = \sum\{y_i \log p_i + (1 - y_i)\log(1 - p_i)\}$ and every term is the log of a number in $(0, 1)$, hence negative.

(c) Take $\beta_0 = 0$, $\beta_1 = c \to \infty$: $p_i \to 0$ for $x_i < 0$ and $p_i \to 1$ for $x_i > 0$, so every term $\to 0$ and $\ell \to 0 = \sup\ell$. By (b) the supremum is not attained, so no maximizer exists (complete separation; the estimates diverge). Remedies: penalized (Firth) likelihood or Bayesian priors.

2025 round (exam of 28 September 2024)

Past problem 9 *[2025 #1]*

(a) The sample sorted, $X_{(1)} \le \cdots \le X_{(n)}$; for a continuous i.i.d. sample, $X_{(k)}$ has density $\frac{n!}{(k-1)!(n-k)!}F^{k-1}(1-F)^{n-k}f$. Used for medians, ranges, extremes, nonparametric statistics. (b) Neyman–Pearson: reject when $L(\theta_1)/L(\theta_0) > k$, with k (and randomization) chosen so the size is exactly α; it maximizes power among all level-α tests. (c) Near-linear dependence among regressors; $X'X$ is nearly singular, so OLS variances inflate, estimates become unstable and may have wrong signs although R^2 is high; diagnosed by VIF, condition number. (d) For fixed n, $n_h \propto N_h S_h$ minimizes $V(\bar{y}_{st})$ (with costs, $n_h \propto N_h S_h/\sqrt{c_h}$): larger and more variable strata get more units. (e) AO affects only one observation ($y_t = x_t + \omega I_t^{(T)}$), like a recording error; IO is a shock to the innovation ($\epsilon_T + \omega$), whose effect propagates through the dynamics as $\omega\psi_j$ at $T+j$. They are distinguished by the pattern of residuals around T and treated differently in outlier-robust estimation.

Past problem 10 *[2025 #2]*

(a) $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$ with sum-to-zero constraints, $\epsilon \overset{\text{iid}}{\sim} N(0,\sigma^2)$. Cell means $\begin{pmatrix} 3 & 2 & 11 \\ 5 & 8 & 1 \end{pmatrix}$, grand mean 5. $S_A = 1.33$ (1), $S_B = 8.0$ (2), $S_{AB} = 138.67$ (2), $S_E = 18$ (6), $S_T = 166$ (11). $F_{AB} = 69.33/3 = 23.1 > F_{0.05}(2,6) = 5.14$: strong interaction. ($F_A = 0.44$, $F_B = 1.33$, not significant.)

(b) A_1B_3, mean 11: $11 \pm t_{0.025}(6)\sqrt{V_E/r} = 11 \pm 2.447\sqrt{3/2} = 11 \pm 3.00 = (8.00, 14.00)$.

(c) Because the interaction is significant, the best level of B depends on A (B_3 is best for A_1 but worst for A_2); combining separately estimated best levels of A and B could pick a poor cell. Estimate the cell means $\mu+\alpha_i+\beta_j+(\alpha\beta)_{ij}$ (option ii). Without interaction, (i) would be preferable since it uses more data per estimate.

Past problem 11 *[2025 #3]*

(a) $\ell = n\log\theta + n\theta\log 2 - (\theta+1)\sum\log x_i$, $\ell' = \frac{n}{\theta} - \sum\log(x_i/2) = 0$: $\hat{\theta} = n/\sum\log(X_i/2)$. $I(\theta) = -E\ell''/n = 1/\theta^2$, so $\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} N(0,\theta^2)$.

(b) $g'(\theta) = -1/(\theta-1)^2$; delta method: $N\big(0, \theta^2/(\theta-1)^4\big)$ (for $\theta \neq 1$).

(c) Second-order expansion: $n\{g(\hat{\theta}) - g(\theta)\} = \frac{1}{2}g''(\theta)\{\sqrt{n}(\hat{\theta}-\theta)\}^2 + o_p(1) \xrightarrow{d} \frac{1}{2}g''(\theta)\theta^2\chi_1^2$, since $\sqrt{n}(\hat{\theta}-\theta)/\theta \xrightarrow{d} Z$ and $Z^2 \sim \chi_1^2$.

Past problem 12 *[2025 #4]*

(a) $M(t) = \sum_x e^{tx}p(x)$; when M is finite near 0 the series may be differentiated term by term: $M^{(n)}(t) = \sum x^n e^{tx}p(x)$, and $t=0$ gives EX^n.

(b) $M(t) = \sum_k \binom{n}{k}(pe^t)^k(1-p)^{n-k} = (pe^t + 1 - p)^n$.

(c) $M' = npe^t(\cdot)^{n-1}$, $M'' = npe^t(\cdot)^{n-1} + n(n-1)p^2e^{2t}(\cdot)^{n-2}$; $EX = np$, $EX^2 = np + n(n-1)p^2$, $\operatorname{Var} X = np(1-p)$.

Past problem 13 *[2025 #5]*

(a) $p(y_1,\dots,y_n) = p(y_1)\prod_{t=2}^{n} p(y_t \mid y_{t-1},\dots,y_1) = \prod_{t=1}^{n} p(y_t \mid y_{t-1})$ by the Markov property (with $y_0 = 0$).

(b) $y_t = \sum_{s=1}^{t} \alpha^{t-s}\epsilon_s$: $Ey_t = 0$, $\operatorname{Var} y_t = \sigma^2 \frac{1-\alpha^{2t}}{1-\alpha^2}$, tending to the stationary value $\sigma^2/(1-\alpha^2)$.

(c) $y_t \mid y_{t-1} \sim N(\alpha y_{t-1}, \sigma^2)$: $p = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\{-\frac{(y_t-\alpha y_{t-1})^2}{2\sigma^2}\}$.

(d) $\ell = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=1}^{n}(y_t - \alpha y_{t-1})^2$; its maximizer is OLS of y_t on y_{t-1}, $\hat\sigma^2$ = mean squared residual.

Past problem 14 *[2025 #6]*

(a) $\sum \lambda_j = \operatorname{tr} R = p$. If regressors were orthogonal, all $\lambda_j = 1$; small eigenvalues ($\lambda_j \approx 0$) indicate near-linear dependencies (the eigenvector gives the combination). Measures: condition number $\sqrt{\lambda_{\max}/\lambda_{\min}}$ (e.g. > 30 serious) or $\sum 1/\lambda_j = \sum \mathrm{VIF}_j$.

(b) $X^{*\prime}X^* = V\Lambda V'$ with V orthogonal. Put $C = X^*V$ and $\alpha = V'\beta^*$; since $VV' = I$, $X^*\beta^* = X^*VV'\beta^* = C\alpha$.

(c) $C'C = V'X^{*\prime}X^*V = \Lambda$ (orthogonal columns), so $\hat\alpha = \Lambda^{-1}C'\mathbf{Y}^*$, i.e. $\hat\alpha_j = \mathbf{c}_j'\mathbf{Y}^*/\lambda_j$. PCR keeps the k components with large λ_j (setting the rest of $\hat\alpha$ to 0), giving $\hat\beta^* = V_k\hat\alpha_k$; then unstandardize: $\hat\beta_j = \hat\beta_j^* s_y/s_{x_j}$ (scaling as used) and $\hat\beta_0 = \bar y - \sum_j \hat\beta_j \bar x_j$. This trades a small bias for a large variance reduction.

Past problem 15 *[2025 #7]*

(a) If $|\alpha| < 1$, $X_t = \sum \alpha^j a_{t-j}$ converges in mean square with constant mean and $\gamma_k = \sigma_a^2\alpha^{|k|}/(1-\alpha^2)$: stationary. If $|\alpha| \geq 1$, a stationary solution would need $\operatorname{Var} X_t = \alpha^2 \operatorname{Var} X_{t-1} + \sigma_a^2$, i.e. $\gamma_0(1-\alpha^2) = \sigma_a^2 > 0$, impossible (for $\alpha = 1$, $\operatorname{Var} X_t$ grows linearly).

(b) $(1-\alpha L)(1-\beta L)Z_t = (1-\beta L)a_t + (1-\alpha L)b_t$, an MA(1) on the right. (i) $\alpha = \beta$: cancel the common factor, $(1-0.4L)Z_t = a_t + b_t$: AR(1) = ARIMA(1,0,0), stationary. (ii) ARMA(2,1) = ARIMA(2,0,1), AR roots 1/0.4, 1/0.5 outside the unit circle: stationary. (iii) $(1-L)(1-0.5L)Z_t =$ MA(1): ARIMA(1,1,1), nonstationary (unit root from X).

Past problem 16 *[2025 #8]*

(a) Mean 0; $v_1 = \sigma_\varepsilon^2$ and $v_t = \sigma_\varepsilon^2 \sum_{j=0}^{t-1} \alpha^{2j} = \sigma_\varepsilon^2 \frac{1-\alpha^{2t}}{1-\alpha^2}$.

(b) (y_t, y_{t-1}) is bivariate normal with $\mathrm{Cov} = \alpha v_{t-1}$, $\operatorname{Var} y_{t-1} = v_{t-1} + \sigma_e^2$. For $t = 1$: $y_0 = e_0$ carries no information on s_1, so $E(y_1 \mid y_0) = 0$, $\mathrm{Var} = \sigma_\varepsilon^2 + \sigma_e^2$. For $t > 1$:

$$E(y_t \mid y_{t-1}) = \frac{\alpha v_{t-1}}{v_{t-1}+\sigma_e^2} y_{t-1}, \qquad \operatorname{Var}(y_t \mid y_{t-1}) = v_t + \sigma_e^2 - \frac{\alpha^2 v_{t-1}^2}{v_{t-1}+\sigma_e^2}.$$

(Conditioning on the whole past, as needed for the likelihood, is done by the Kalman filter.)

(c) $L = p(y_0)\prod_{t=1}^{T} p(y_t \mid y_{0:t-1})$ (prediction-error decomposition). With $\hat s_{0|0} = 0$, $P_{0|0} = 0$, for $t \geq 1$: predict $\hat s_{t|t-1} = \alpha \hat s_{t-1|t-1}$, $P_{t|t-1} = \alpha^2 P_{t-1|t-1} + \sigma_\varepsilon^2$; then $y_t \mid y_{0:t-1} \sim N(\hat s_{t|t-1}, F_t)$ with $F_t = P_{t|t-1} + \sigma_e^2$; update (Bayes) $K_t = P_{t|t-1}/F_t$, $\hat s_{t|t} = \hat s_{t|t-1} + K_t(y_t - \hat s_{t|t-1})$, $P_{t|t} = (1-K_t)P_{t|t-1}$. Hence

$$L(\alpha) = \phi(y_0; 0, \sigma_e^2) \prod_{t=1}^{T} \frac{1}{\sqrt{2\pi F_t}} \exp\Big\{ -\frac{(y_t - \hat s_{t|t-1})^2}{2F_t} \Big\},$$

maximized numerically in α.

2024 round (exam of 16 September 2023)

Past problem 17 *[2024 #1]*

(a) A census enumerates every unit (no sampling error, but costly, slow, and with large nonsampling errors in practice); a sample survey observes a probability sample and infers to the population with quantifiable sampling error, allowing better-trained staff and more detailed questions. (b) Start from the full model; repeatedly remove the variable with the largest p-value (smallest partial F) if it exceeds a threshold, refit, and stop when all remaining variables are significant. Simple, but may miss good subsets and inflates significance of the final model. (c) $P = 1/6$, odds $= \frac{1/6}{5/6} = \frac{1}{5} = 0.2$. (d) T is complete for $\{P_\theta\}$ if $E_\theta g(T) = 0$ for all θ implies $g(T) = 0$ a.s.; with sufficiency it yields unique UMVUEs (Lehmann–Scheffé) and independence from ancillaries (Basu). (e) For i.i.d. X_i with mean μ and finite variance σ^2, $\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} N(0,1)$ whatever the distribution: the basis of large-sample tests and intervals.

Past problem 18 *[2024 #2]*

(a) $\sum x^3 = 225$, $k = 1/225$. (b) $(1 + 8)/225 = 0.04$. (c) $P(Y = 10) = 0.6$, $P(Y = 20) = 0.4$; $P(Z = 10, 20, 30) = (0.5, 0.2, 0.3)$. (d) $EY = 14$, $EZ = 18$. Not independent: $P(10, 20) = 0.1 \neq 0.6 \times 0.2 = 0.12$. (e) $(10 \cdot 0.2 + 20 \cdot 0.1 + 30 \cdot 0.1)/0.4 = 17.5$.

Past problem 19 *[2024 #3]*

(a) $EX = \frac{1}{b-a}\int_a^b x\,dx = \frac{a+b}{2}$; $EX^2 = \frac{a^2+ab+b^2}{3}$, so $\operatorname{Var} X = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$.

(b) $\binom{n}{x}p^x(1-p)^{n-x} = \frac{n(n-1)\cdots(n-x+1)}{n^x}\frac{\lambda^x}{x!}\big(1 - \frac{\lambda}{n}\big)^n\big(1 - \frac{\lambda}{n}\big)^{-x} \to 1 \cdot \frac{\lambda^x}{x!}e^{-\lambda} \cdot 1$.

Past problem 20 *[2024 #4]*

(a) $M' = M$; $M^2 = I - 2\frac{\mathbf{i}\mathbf{i}'}{n} + \frac{\mathbf{i}(\mathbf{i}'\mathbf{i})\mathbf{i}'}{n^2} = M$; $M\mathbf{i} = \mathbf{i} - \mathbf{i} = 0$; $\operatorname{rank} M = \operatorname{tr} M = n - 1$ (idempotent).

(b) $\mathbf{i}'\mathbf{y} = \sum y_i$, $\mathbf{i}'\mathbf{i} = n$. $M\mathbf{y} = \mathbf{y} - \bar{y}\mathbf{i}$ (deviations), so $\mathbf{y}'M\mathbf{y} = (M\mathbf{y})'(M\mathbf{y}) = \sum(y_i - \bar{y})^2$.

(c) Choose orthogonal Q with $Q'MQ = \operatorname{diag}(I_{n-1}, 0)$ and set $\mathbf{z} = Q'\mathbf{y}/\sigma \sim N(0, I)$. Then $\mathbf{y}'M\mathbf{y}/\sigma^2 = \mathbf{z}'Q'MQ\mathbf{z} = \sum_{i=1}^{n-1} z_i^2 \sim \chi^2_{n-1}$.

(d) $\sqrt{n}\bar{y}/\sigma \sim N(0,1)$, independent of $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$; the ratio $\frac{\sqrt{n}\bar{y}/\sigma}{\sqrt{s^2/\sigma^2}} = \frac{\bar{y}}{\sqrt{s^2/n}}$ is t_{n-1} by definition.

Past problem 21 *[2024 #5]*

(a) $EY = \int_2^\infty 24y^{-3}dy = 3$, $EY^2 = \int_2^\infty 24y^{-2}dy = 12$, $\operatorname{Var} Y = 3$. $\sum Y \approx N(144, 144)$: $P(\sum Y > 180) \approx 1 - \Phi(3) = 0.0013$.

(b) $P(Z_{(n)} \le z) = (z/2\theta)^n$, $EZ_{(n)} = \frac{n}{n+1}2\theta$; unbiased: $\frac{n+1}{2n}Z_{(n)}$.

(c) $\hat{\beta} = \sum x_i y_i / \sum x_i^2$; $E(\hat{\beta} \mid X) = \beta$; $\operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 / \sum x_i^2$.

Past problem 22 *[2024 #6]*

(a) $\ell = n\log\theta - \theta\sum x_i$, $\ell' = 0$ at $\hat\theta = 1/\bar x$; here $1/2 = 0.5$.

(b) Exponential with mean 1/3: $\pi(\theta) = 3e^{-3\theta}$, $\theta > 0$.

(c) $\pi(\theta \mid \mathbf{x}) \propto \theta^3 e^{-6\theta} e^{-3\theta} = \theta^3 e^{-9\theta}$: Gamma(shape 4, rate 9), density $\frac{9^4}{3!}\theta^3 e^{-9\theta}$.

(d) $E(\theta \mid \mathbf{x}) = 4/9 = 0.444$. It shrinks the MLE 0.5 toward the prior mean 0.333; the weight on the prior vanishes as n grows. (The posterior mode is $3/9 = 0.333$.)

Past problem 23 *[2024 #7]*

(a) $x_{ij} = \mu + a_i + b_j + e_{ij}$, $\sum a_i = \sum b_j = 0$, $e_{ij} \overset{\text{iid}}{\sim} N(0,\sigma^2)$; $H_0^A : a_i = 0$ for all i, $H_0^B : b_j = 0$ for all j.

(b) $(\bar x_{i\cdot} - \bar{\bar x})$: variation due to A; $(\bar x_{\cdot j} - \bar{\bar x})$: due to B; $(x_{ij} - \bar x_{i\cdot} - \bar x_{\cdot j} + \bar{\bar x})$: residual (error, including any interaction). Cross products vanish.

(c)–(d) A means $10, 10.33, 6.67$; B means $10, 7.67, 9.33$; grand mean 9. $S_A = 24.67$ (2), $S_B = 8.67$ (2), $S_E = 74.67$ (4), $S_T = 108$ (8). $F_A = 12.33/18.67 = 0.66$, $F_B = 4.33/18.67 = 0.23$, both $< F_{0.05}(2,4) = 6.94$: no significant differences.

Past problem 24 *[2024 #8]*

(a) Ratio estimator $\hat r = \sum y_i / \sum m_i = 11990/129 = 92.95$. With $s_r^2 = \frac{1}{n-1}\sum(y_i - \hat r m_i)^2 = 181{,}328$ and $\bar m = 6.45$: $\hat V(\hat r) = \frac{1-n/N}{n\bar m^2} s_r^2 = \frac{0.9333}{20(41.6)}(181328) = 203.4$ (se 14.3).

(b) $n = \frac{N s_r^2}{ND + s_r^2}$ with $D = B^2\bar m^2/4 = 26{,}000$: $n = 6.8 \approx 7$ clusters.

(c) Stratum 2: $\hat r_2 = 3800/43 = 88.37$, $\hat V_2 = 597.7$. Weight by estimated population sizes $\hat M_1 = 300(6.45) = 1935$, $\hat M_2 = 200(4.3) = 860$: $W = (0.692, 0.308)$. Estimate $0.692(92.95) + 0.308(88.37) = 91.54$; variance $0.692^2(203.4) + 0.308^2(597.7) = 154.1$.

(d) Stratified cluster sampling (clusters sampled within each city-stratum, ratio estimation in each, combined with population weights). The combined estimate is lower (stratum 2 is poorer) and its variance is smaller in absolute terms because the second, independently sampled stratum adds information; but it estimates a different population (both cities).

Past problem 25 *[2024 #9]*

(a) Spurious regression: OLS between independent $I(1)$ series gives significant t statistics and high R^2 with no real relation. Solutions: difference the series (losing long-run information) or, if they are cointegrated, estimate the long-run relation and an error-correction model.

(b) Engle–Granger: estimate $y_t = a + bx_t + \epsilon_t$ by OLS and apply an ADF test to the residuals, using Engle–Granger (not Dickey–Fuller) critical values because b is estimated; first confirm each series is $I(1)$. Johansen's trace/max-eigenvalue tests handle more than two variables and give the number of cointegrating vectors. Caveats: low power, sensitivity to lag length, deterministic terms and structural breaks; the normalization in EG is arbitrary.

(c) Subtract y_{t-1} and write $2x_t = 2\Delta x_t + 2x_{t-1}$: $\Delta y_t = 4 + 2\Delta x_t - 0.5y_{t-1} + (2+\theta)x_{t-1} + u_t = 4 + 2\Delta x_t - 0.5\{y_{t-1} - 2(2+\theta)x_{t-1}\} + u_t$. Matching the cointegrating coefficient $2(2+\theta) = 5$ gives $\theta = 0.5$, and the ECM is $\Delta y_t = 4 + 2\Delta x_t - 0.5(y_{t-1} - 5x_{t-1}) + u_t$: half of any disequilibrium is corrected each period.

2023 round (exam of 24 September 2022)

Past problem 26 *[2023 #1]*

(a) Pearson correlation of the ranks; $r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$ without ties. It measures monotone association and is robust to outliers and nonlinearity. (b) In $X_j = \sum_k l_{jk}F_k + \epsilon_j$, $\operatorname{Var} X_j = h_j^2 + \psi_j$ with communality $h_j^2 = \sum_k l_{jk}^2$: the share of variance explained by the common factors; ψ_j is the specific variance. (c) In $\mathbf{y} = X\beta + \epsilon$ with $E\epsilon = 0$, $\operatorname{Var}\epsilon = \sigma^2 I$, OLS is the best linear unbiased estimator: for any linear unbiased $\tilde{\beta}$, $\operatorname{Var}\tilde{\beta} - \operatorname{Var}\hat{\beta}$ is positive semidefinite. Normality is not required. (d) $\mathrm{VIF}_j = 1/(1-R_j^2)$, the factor by which multicollinearity inflates $\operatorname{Var}\hat{\beta}_j$; values above 10 signal trouble. (e) $\epsilon_t = \sigma_t\eta_t$, $\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q}\alpha_i\epsilon_{t-i}^2$: conditional variance depending on past squared shocks, capturing volatility clustering and fat tails in financial returns (Engle 1982).

Past problem 27 *[2023 #2]*

(a) Under H_0, $Y_i \in \{0,1\}$; under H_1, $Y_i \in \{1,2\}$. Reject H_0 iff $\max Y_i = 2$ (some $Y_i = 2$). This never happens under H_0, so the type I error is 0. (Equally, $\min Y_i = 0$ proves H_0.)

(b) Power $= P_1(\text{some } Y_i = 2) = 1 - (\frac{1}{2})^n$.

Past problem 28 *[2023 #3]*

(a) $y_{ij} = \mu + a_i + e_{ij}$, $\sum a_i = 0$, $e_{ij} \overset{\text{iid}}{\sim} N(0,\sigma_E^2)$. Means $11.67, 10, 8.33, 8$; grand mean 9.5. $S_A = 25.67$ (3), $S_E = 5.83$ (8), $S_T = 31.5$ (11); $V_A = 8.56$, $V_E = 0.729$, $F = 11.73$.

(b) $F > F_{0.05}(3,8) = 4.07$: significant. $\mu(A_3)$: $8.33 \pm t_{0.05}(8)\sqrt{0.729/3} = 8.33 \pm 1.860(0.493) = 8.33 \pm 0.92 = (7.42, 9.25)$.

(c) $\left(\frac{S_E}{\chi^2_{0.025}(8)}, \frac{S_E}{\chi^2_{0.975}(8)}\right) = \left(\frac{5.83}{17.53}, \frac{5.83}{2.18}\right) = (0.33, 2.68)$.

(d) Randomized block design with a random block factor (mixed model): $y_{ij} = \mu + a_i + b_j + e_{ij}$, a_i fixed with $\sum a_i = 0$, $b_j \overset{\text{iid}}{\sim} N(0,\sigma_B^2)$ independent of e_{ij}; randomization within each day. Day-to-day variation is removed from the error, increasing precision; one can estimate σ_B^2 and generalize to days in general. The first experiment is a completely randomized one-way design without blocking.

Past problem 29 *[2023 #4]*

(a) Means $99, 100, 98, 100$; $W_h = (0.2, 0.3, 0.25, 0.25)$; $\bar{y}_{st} = 99.3$. $\hat{V} = \sum W_h^2(1 - \frac{n_h}{N_h})\frac{s_h^2}{n_h} = 0.04(16.39) + 0.09(7.72) + 0.0625(6.58) + 0.0625(18.75) = 2.93$.

(b) $\bar{y} = 1986/20 = 99.3$, $s^2 = 59.8$, $\hat{V} = (1 - \frac{20}{120})\frac{59.8}{20} = 2.49$.

(c) The estimates coincide here, and stratification does not reduce the variance: the chain means are nearly equal (little between-strata variation), and the allocation is not optimal (chain D, the most variable, is not sampled more heavily; chain A is oversampled relative to proportional allocation). Stratification gains when strata differ in mean; it also guarantees coverage of each chain and gives chain-level estimates.

Past problem 30 *[2023 #5]*

(a) $\bar{X}$ and $\frac{1}{n}\sum X_i^2 - \bar{X}^2 = \hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$. (b) The same: $\ell = -\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum(x_i-\mu)^2$ gives $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$. (c) $\bar{X}$ is unbiased and consistent (WLLN). $E\hat{\sigma}^2 = \frac{n-1}{n}\sigma^2$: biased, asymptotically unbiased; $\operatorname{Var}\hat{\sigma}^2 = \frac{2(n-1)\sigma^4}{n^2} \to 0$, so consistent. $S^2 = \frac{n}{n-1}\hat{\sigma}^2$ is unbiased.

Past problem 31 *[2023 #6]*

(a) $1-0.8z+0.1z^2 = 0$ has roots $z = 1.55$ and 6.45, outside the unit circle: stationary. (Equivalently $|\phi_2| < 1$, $\phi_1+\phi_2 < 1$, $\phi_2-\phi_1 < 1$.) $\dot{X}_t = \sum \psi_j \epsilon_{t-j}$ with $\psi_0 = 1$, $\psi_1 = 0.8$, $\psi_j = 0.8\psi_{j-1} - 0.1\psi_{j-2}$: $0.54, 0.352, 0.2276, \ldots$; in closed form $\psi_j = \frac{\lambda_1^{j+1} - \lambda_2^{j+1}}{\lambda_1 - \lambda_2}$ with $\lambda_{1,2} = 0.4 \pm \sqrt{0.06} = 0.645, 0.155$.

(b) Yule–Walker: $\rho_1 = \phi_1/(1-\phi_2) = 0.8/1.1 = 0.727$, $\rho_2 = \phi_1\rho_1 + \phi_2 = 0.582 - 0.1 = 0.482$.

Past problem 32 *[2023 #7]*

(a) $(X'X)^{-1} = \frac{1}{0.36}\begin{pmatrix} 1 & -0.8 \\ -0.8 & 1 \end{pmatrix} = \begin{pmatrix} 2.78 & -2.22 \\ -2.22 & 2.78 \end{pmatrix}$.

(b) Setting the gradient $-2X'(\mathbf{y} - X\beta) + 2\lambda\beta$ to 0: $\hat{\beta}^\lambda = (X'X + \lambda I)^{-1}X'\mathbf{y}$.

(c) $E\hat{\beta}^\lambda = (X'X + \lambda I)^{-1}X'X\beta$, so bias $= -\lambda(X'X + \lambda I)^{-1}\beta \neq 0$ for $\beta \neq 0$.

(d) $\operatorname{Var}\hat{\beta}^\lambda = \sigma^2(X'X + \lambda I)^{-1}X'X(X'X + \lambda I)^{-1} = \begin{pmatrix} 0.85 & -0.40 \\ -0.40 & 0.85 \end{pmatrix}$: variances drop from 2.78 to 0.85. With correlated regressors, ridge accepts a small bias for a large variance reduction, and for suitable λ has smaller MSE than OLS (Hoerl–Kennard). In eigen terms, the components along small eigenvalues (0.2 here) are shrunk most.

Past problem 33 *[2023 #8]*

(a) $f = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\{-\frac{1}{2(1-\rho^2)}(z_1^2 - 2\rho z_1 z_2 + z_2^2)\}$, $z_i = (x_i - \mu_i)/\sigma_i$.

(b) $f(x_1, x_2)/f_1(x_1)$: since $z_1^2 - 2\rho z_1 z_2 + z_2^2 = (1-\rho^2)z_1^2 + (z_2 - \rho z_1)^2$, the ratio is $\propto \exp\{-\frac{(z_2 - \rho z_1)^2}{2(1-\rho^2)}\}$, i.e. normal with the stated mean and variance.

(c) $\rho = 12/20 = 0.6$; mean $1 + 0.6 \cdot 5 \cdot (4/4) = 4$, variance $25(0.64) = 16$. $P = \Phi(\frac{12-4}{4}) - \Phi(\frac{5-4}{4}) = \Phi(2) - \Phi(0.25) = 0.9772 - 0.5987 = 0.379$.

(d) $\max(X_1, X_2) = \frac{X_1+X_2}{2} + \frac{|X_1 - X_2|}{2}$ and $X_1 - X_2 \sim N(0, 2)$, $E|N(0,2)| = \sqrt{2}\sqrt{2/\pi}$, so $E\max = \frac{1}{\sqrt{\pi}} = 0.564$.

Past problem 34 *[2023 #9 (essay)]*

Outline. (1) Diversity: new data sources capture phenomena traditional surveys miss and add timeliness (nowcasting); demand from policy in fast-changing conditions. (2) Stability: an indicator is valuable only if its relation to the target is stable; proxies based on behaviour (lipstick sales) break when behaviour changes for unrelated reasons (masks), i.e. a structural break in the measurement model; official statistics carry authority, so errors damage trust. (3) Balance via process: publish new statistics first as experimental statistics with clear labels; validate against established benchmarks, test for structural stability (rolling-window correlations, Chow/CUSUM tests, out-of-sample tracking), monitor coverage and representativeness, document methodology and uncertainty; graduate to official status after a track record. (4) Position (example): stability as the precondition for official statistics, diversity pursued through an experimental tier — not a choice of one over the other.

2022 round (exam of 11 September 2021)

Past problem 35 *[2022 #1]*

(a) $\hat{Y} = X\hat{\beta} = X(X'X)^{-1}X'Y$, residual $e = M_X Y$; M_X is symmetric idempotent, so $SSE = e'e = Y'M_X Y$.

(b) Take $P = M_X/\sigma^2$, $\Sigma = \sigma^2 I$: $P\Sigma = M_X$ is idempotent with rank $\operatorname{tr} M_X = n - p$, and the noncentrality $\frac{1}{2\sigma^2}\beta' X' M_X X \beta = 0$ since $M_X X = 0$. So $SSE/\sigma^2 \sim \chi^2_{n-p}$, $E(SSE) = (n-p)\sigma^2$, $E(MSE) = \sigma^2$.

(c) $F = \frac{(672-480)/2}{480/20} = \frac{96}{24} = 4.0 > F_{0.05}(2, 20) = 3.49$: reject H_0; $\beta_2 \neq 0$.

Past problem 36 *[2022 #2]*

(a) Expected 20 each; $\chi^2 = \frac{4+9+16+1+4+16}{20} = 2.5 < \chi^2_{0.05}(5) = 11.07$: do not reject fairness.

(b) $P(\geq 5) = \frac{1}{3}$. Heads: $P(\mathrm{Bin}(6, \frac{1}{3}) \geq 4) = \frac{15\cdot 4 + 6\cdot 2 + 1}{729} = \frac{73}{729} = 0.1001$. Tails: $P(\mathrm{Bin}(6, \frac{2}{3}) \geq 5) = \frac{6\cdot 32 + 64}{729} = \frac{256}{729} = 0.3512$. Win: $\frac{1}{2}(0.1001 + 0.3512) = \frac{329}{1458} = 0.2257$.

(c) See 2024 #3(b).

Past problem 37 *[2022 #3]*

(a) With inclusion indicators Z_j, $\bar{y} = \frac{1}{n}\sum Z_j Y_j$, $P(Z_j = 1) = \frac{n}{N}$, $\operatorname{Var} Z_j = \frac{n}{N}(1 - \frac{n}{N})$, $\operatorname{Cov}(Z_j, Z_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}$. Using $\sum_{j\neq k}(Y_j - \mu)(Y_k - \mu) = -\sum(Y_j - \mu)^2$: $\operatorname{Var}\bar{y} = \frac{1}{n^2}\{\frac{n(N-n)}{N^2}\sum(Y_j - \mu)^2 + \frac{n(N-n)}{N^2(N-1)}\sum(Y_j - \mu)^2\} = \frac{\sigma^2}{n}\frac{N-n}{N-1}$.

(b) $Es^2 = \frac{N}{N-1}\sigma^2 = S^2$ (by a similar computation), so $E\{\frac{N-n}{N}\frac{s^2}{n}\} = \frac{N-n}{N}\frac{N\sigma^2}{(N-1)n} = \operatorname{Var}\bar{y}$.

(c) $\hat{\mu} = 320$; $\widehat{\mathrm{se}} = \sqrt{0.9 \cdot 1470/300} = 2.1$; bound 4.2.

(d) Quintiles: $W_h = 0.2$, $\sum W_h s_h^2 = 1080$; $\widehat{\mathrm{se}} = \sqrt{(1-f)\sum W_h s_h^2/n} = \sqrt{0.9(1080)/300} = 1.8$, smaller than 2.1 because between-quintile variation is removed.

Past problem 38 *[2022 #4]*

(a) For $\lambda_2 > \lambda_1$, $\frac{L(\lambda_2)}{L(\lambda_1)} = e^{-n(\lambda_2 - \lambda_1)}(\lambda_2/\lambda_1)^{\sum x_i}$ is increasing in $T = \sum x_i$.

(b) A level-α test ϕ is UMP if $E_\lambda \phi \geq E_\lambda \phi'$ for all $\lambda \in \Theta_1$ and every level-α test ϕ'. By Karlin–Rubin, the test rejecting for $T > k$, randomizing with probability γ at $T = k$, with $P_{\lambda_0}(T > k) + \gamma P_{\lambda_0}(T = k) = \alpha$ and $T \sim \mathrm{Poisson}(n\lambda_0)$, is UMP.

Past problem 39 *[2022 #5]*

(a) $\hat{\beta} = (X'X)^{-1}X'Y \sim N(\beta, \sigma^2(X'X)^{-1})$. Any linear unbiased $\tilde{\beta} = CY$ needs $CX = I$; write $C = (X'X)^{-1}X' + D$ with $DX = 0$. Then $\operatorname{Var}\tilde{\beta} = \sigma^2\{(X'X)^{-1} + DD'\}$, exceeding $\operatorname{Var}\hat{\beta}$ by the p.s.d. $\sigma^2 DD'$ (Gauss–Markov).

(b) OLS is still unbiased but not BLUE; $\operatorname{Var}\hat{\beta}_{\mathrm{OLS}} = (X'X)^{-1}X'VX(X'X)^{-1}$. The BLUE is GLS/WLS: $\hat{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}Y$, i.e. weighted LS with weights $1/\lambda_i$, with variance $(X'V^{-1}X)^{-1}$ (apply (a) to $V^{-1/2}Y$).

Past problem 40 *[2022 #6]*

(a) A: $x_0 - \ell'(x_0)/\ell''(x_0)$; B: $i+1$; C: x_1.

(b) Resample the data with replacement many times, recompute the statistic each time, and use the empirical distribution of the replicates to estimate its standard error, bias or confidence interval.

(c) Spectral (eigen) decomposition. $Y \sim N(0,\Lambda)$: components are independent, $\operatorname{Var} Y_i = \lambda_i$ in decreasing order, $\sum \lambda_i = \operatorname{tr}\Sigma = \sum \sigma_i^2$, so $\lambda_k / \sum \lambda_i$ is the proportion of variance explained by the k-th component.

(d) Choose k initial centroids (e.g. random data points); repeat: assign each point to the nearest centroid; recompute each centroid as the mean of its points; until assignments no longer change (or a maximum number of iterations). Output the k clusters. (Run from several starts; the result is a local optimum of the within-cluster sum of squares.)

(e) An unknown value is set equal to the value at the nearest observed point (in time or space). It is simple and preserves observed values, but produces a step function: discontinuities, no use of other neighbours or trends, poor accuracy where points are sparse.

Past problem 41 *[2022 #7]*

(a) $\pi(p \mid \mathbf{y}) \propto p^{\sum y_i+\alpha-1}(1-p)^{n-\sum y_i+\beta-1}$: $\text{Beta}(\alpha+\sum y_i, \beta+n-\sum y_i)$.

(b) $\frac{\alpha+\sum y_i}{\alpha+\beta+n}$, a weighted average of the prior mean $\frac{\alpha}{\alpha+\beta}$ and $\bar{y}$.

(c) $\rho(a) = E\{(a-p)^2 \mid \mathbf{y}\} = a^2 - 2aE(p \mid \mathbf{y}) + E(p^2 \mid \mathbf{y})$; $\rho'(a) = 2a - 2E(p \mid \mathbf{y}) = 0$ and $\rho'' = 2 > 0$.

Past problem 42 *[2022 #8]*

(a) $L = (2\theta)^{-n}\mathbf{1}\{\theta > \max|x_i|\}$: $\hat{\theta}_n = \max|X_i|$.

(b) $|X_i| \sim U(0,\theta)$, so $P(\theta - \hat{\theta}_n \le t) = 1 - (1 - t/\theta)^n$ for $0 \le t \le \theta$ (0 below, 1 above).

(c) $C_n = n$: $P(n(\theta - \hat{\theta}_n) \le t) = 1 - (1 - \frac{t}{n\theta})^n \to 1 - e^{-t/\theta}$, the exponential with mean θ.

Past problem 43 *[2022 #9]*

(a) $\rho_k = \phi^k$. (b) $\rho_1 = -\theta/(1+\theta^2)$, $\rho_k = 0$ for $k \ge 2$.

(c) With γ_k for the mean-corrected series: $\gamma_0 = \phi\gamma_1 + \sigma^2(1 - \theta(\phi - \theta))$, $\gamma_1 = \phi\gamma_0 - \theta\sigma^2$, giving

$$\rho_1 = \frac{(1-\phi\theta)(\phi-\theta)}{1+\theta^2-2\phi\theta}, \qquad \rho_k = \phi\rho_{k-1} \ (k \ge 2).$$

(d) See Part I, MA(1)/ARMA problem (d): AR(p) has PACF cutting off after p, MA(q) has ACF cutting off after q, ARMA has both tailing off. Limitations: sample ACF/PACF are noisy and correlated, mixed models are hard to read, several models fit similarly. Use AIC $= -2\log L + 2k$, BIC $= -2\log L + k\log n$ over candidate (p,q), choose the minimum, and check residuals with the Ljung–Box test; prefer parsimony.

Past problem 44 *[2022 #10 (essay)]*

Outline. (1) Strengths of ML: flexible nonlinear prediction with many predictors (nowcasting GDP with high-dimensional data, text analysis of central bank communications, anomaly detection in

payment systems, early-warning credit risk). (2) Strengths of statistical models: interpretability, causal and structural inference (policy counterfactuals), uncertainty quantification, stability over time. (3) Risks: overfitting and instability under regime change (Lucas critique), black-box decisions in supervision, bias in training data, privacy of micro data (anonymization, differential privacy), reproducibility. (4) Approach: use ML for prediction tasks with out-of-sample validation, statistical models for inference and policy analysis; combine (e.g. double/debiased ML for causal effects, interpretable ML via SHAP); governance for data quality and model risk. (5) View: complementary tools; the choice follows the question (prediction vs explanation).

2021 round (exam of 12 September 2020)

Past problem 45 *[2021 #1]*

(a) $P(X = 0, 2, 4) = (0.3, 0.5, 0.2)$; $P(Y = 1, 2, 3) = (\frac{8}{30}, \frac{12}{30}, \frac{10}{30})$. (b) $EX = 1.8$, $EX^2 = 5.2$, $\operatorname{Var} X = 1.96$, $EY = \frac{62}{30} = 2.067$, $E(X + Y) = 3.867$. (c) No: $P(X = 4, Y = 3) = 0 \neq 0.2 \cdot \frac{1}{3}$.

Past problem 46 *[2021 #2]*

(a) $\{|X_n + Y_n - X - Y| \geq \epsilon\} \subset \{|X_n - X| \geq \epsilon/2\} \cup \{|Y_n - Y| \geq \epsilon/2\}$; both probabilities $\to 0$. For $a \neq 0$, $P(|aX_n - aX| \geq \epsilon) = P(|X_n - X| \geq \epsilon/|a|) \to 0$.

(b) Given ϵ, choose δ with $|x - a| < \delta \Rightarrow |g(x) - g(a)| < \epsilon$. Then $P(|g(X_n) - g(a)| \geq \epsilon) \leq P(|X_n - a| \geq \delta) \to 0$.

(c) Markov's inequality with $g = (\bar{X} - \mu)^2$: $P(|\bar{X} - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \to 0$.

(d) Consistent.

Past problem 47 *[2021 #3]*

(a) $E_i = -\log(1 - F(X_i))$ are i.i.d. Exp(1) and $X_i = h(E_i)$ with h increasing, so $X_{(r)} = h(E_{(r)})$. By memorylessness the spacings $E_{(r)} - E_{(r-1)}$ are independent Exp with rate $n - r + 1$ (Part I, order-statistics problem), i.e. $E_{(r)} \overset{d}{=} \sum_{k=1}^{r} \frac{Z_k}{n-k+1}$ jointly in r (Rényi's representation).

(b) $\sum_{i<j}(X_{(j)} - X_{(i)}) = \sum_i (2i - n - 1)X_{(i)}$ ($X_{(i)}$ is larger than $i-1$ and smaller than $n-i$ others). With $\binom{10}{2} = 45$: $a_i = \frac{2i-11}{45}$, i.e. $-\frac{9}{45}, -\frac{7}{45}, \ldots, \frac{9}{45}$.

(c) $E|X - \mu| = 2\sigma \int_0^\infty z\phi(z)dz = 2\sigma \frac{1}{\sqrt{2\pi}} = \sigma\sqrt{2/\pi}$.

(d) $EG_n = E|X_1 - X_2|$ and $X_1 - X_2 \sim N(0, 2\sigma^2)$: $\sqrt{2}\sigma\sqrt{2/\pi} = 2\sigma/\sqrt{\pi}$.

Past problem 48 *[2021 #4]*

(a) $y_{ij} = \mu + a_i + e_{ij}$, $\sum a_i = 0$, $e_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(b) $SS_T = \sum\sum(y_{ij} - \bar{y})^2 = \sum\sum(\bar{y}_i - \bar{y})^2 + \sum\sum(y_{ij} - \bar{y}_i)^2$, cross term zero. Means $64, 57.33, 45.33, 56.33$; grand mean 55.75.

(c) $SS_A = 538.25$ (3), $SS_E = 202.0$ (8), $SS_T = 740.25$ (11); $MS_A = 179.42$, $MS_E = 25.25$, $F = 7.11 > 4.07$: significant.

(d) Half-width $t_{0.025}(8)\sqrt{25.25/3} = 2.306(2.90) = 6.69$: ① $(57.31, 70.69)$, ③ $(38.64, 52.02)$. Difference $18.67 \pm 2.306\sqrt{2(25.25)/3} = 18.67 \pm 9.46 = (9.21, 28.13)$.

Past problem 49 *[2021 #5]*

(a) $\gamma_0 = 1 + 2.4^2 + 0.8^2 = 7.4$, $\gamma_1 = 2.4 + 2.4(0.8) = 4.32$, $\gamma_2 = 0.8$, $\gamma_k = 0$ for $k \geq 3$.

(b) $1 + 2.4z + 0.8z^2 = (1 + 2z)(1 + 0.4z)$ has roots -0.5 and -2.5; the root -0.5 lies inside the unit circle.

(c) Replace the factor $(1 + 2B)$ by $(1 + 0.5B)$ and multiply the noise variance by 2^2: $Z_t = (1 + 0.5B)(1 + 0.4B)\eta_t = \eta_t + 0.9\eta_{t-1} + 0.2\eta_{t-2}$, $\eta_t \sim WN(0, 4)$. Check: $\gamma_0 = 4(1.85) = 7.4$, $\gamma_1 = 4(0.9 + 0.18) = 4.32$, $\gamma_2 = 0.8$.

Past problem 50 *[2021 #6]*

(a) $C\hat{\beta} - m \sim N(C\beta - m, \sigma^2 C(X'X)^{-1}C')$, independent of s^2, so

$$F = \frac{(C\hat{\beta} - m)'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - m)/q}{s^2} \sim F(q, n - p) \text{ under } H_0$$

(equivalently $\frac{(SSE_R - SSE_F)/q}{SSE_F/(n-p)}$).

(b) $C = e_j'$, $q = 1$: $F = \hat{\beta}_j^2/\{s^2[(X'X)^{-1}]_{jj}\} = t^2$, and $t^2 \sim F(1, n - p)$ when $t \sim t_{n-p}$; the critical values satisfy $F_\alpha(1, \nu) = t_{\alpha/2}(\nu)^2$, so the decisions coincide.

Past problem 51 *[2021 #7]*

(a) $L = (2\pi)^{-(n+m)p/2}|\Sigma|^{-(n+m)/2} \exp\{-\frac{1}{2} \operatorname{tr} \Sigma^{-1}[\sum(x_i - \mu_x)(x_i - \mu_x)' + \sum(y_j - \mu_y)(y_j - \mu_y)']\}$.

(b) $\hat{\mu}_0 = \frac{n\bar{x} + m\bar{y}}{n+m}$ (the pooled mean), $\hat{\Sigma}_0 = \frac{1}{n+m}\{\sum(x_i - \hat{\mu}_0)(x_i - \hat{\mu}_0)' + \sum(y_j - \hat{\mu}_0)(y_j - \hat{\mu}_0)'\}$.

(c) $\hat{\mu}_x = \bar{x}$, $\hat{\mu}_y = \bar{y}$, $\hat{\Sigma} = \frac{(n-1)S_x + (m-1)S_y}{n+m} = \frac{n+m-2}{n+m}S$. (The LRT of $\mu_x = \mu_y$ based on these is equivalent to Hotelling's T^2.)

Past problem 52 *[2021 #8 (essay)]*

Outline. (1) New high-frequency indicators: card spending, mobility, electricity use, real-time payment and tax data, web-scraped prices; weekly economic activity indices (e.g. dynamic factor models). Timely but with coverage bias and short histories. (2) Improved methods: flash estimates and nowcasts with mixed-frequency models (MIDAS, bridge equations), more administrative data to replace delayed surveys, outlier/level-shift treatment in seasonal adjustment during the shock (so that seasonal factors are not distorted), imputation for survey nonresponse during lockdowns. (3) Evaluation: timeliness gains vs accuracy losses measured by revisions; publish uncertainty and methodology; maintain continuity of core statistics. (4) Lessons: invest in data infrastructure and partnerships beforehand.

2019 round (exam of 20 October 2018)

Past problem 53 *[2019 #1]*

(a) $CV = \sigma/\mu$ (sample $s/\bar{x}$), a unit-free measure of relative dispersion for comparing variables on different scales (meaningful for positive variables). (b) For convex g, $g(EX) \leq Eg(X)$ (reversed

for concave g); e.g. $(EX)^2 \le EX^2$, $E\log X \le \log EX$. (c) $F(x) = 1/(1+e^{-(x-\mu)/s})$, symmetric, mean μ, variance $\pi^2 s^2/3$, heavier tails than the normal; its cdf is the link of logistic regression. (d) $P(X > s+t \mid X > s) = P(X > t)$; the exponential is the only continuous distribution with it (geometric for discrete); constant hazard. (e) $I(\theta) = E\{(\partial \log f/\partial\theta)^2\} = -E\{\partial^2 \log f/\partial\theta^2\}$ under regularity; it measures the information about θ in one observation, gives the Cramér–Rao bound $1/\{nI(\theta)\}$ and the asymptotic variance of the MLE.

Past problem 54 *[2019 #2]*

(a) $\int\{\int x_2 f(x_2 \mid x_1)dx_2\}f_1(x_1)dx_1 = \int\int x_2 f(x_1,x_2)dx_2dx_1 = EX_2$.

(b) $\operatorname{Var} X_2 = E\{\operatorname{Var}(X_2 \mid X_1)\} + \operatorname{Var}\{E(X_2 \mid X_1)\}$: expand $E(X_2-\mu)^2$ with $X_2-\mu = \{X_2 - E(X_2 \mid X_1)\} + \{E(X_2 \mid X_1)-\mu\}$; the cross term has zero expectation by (a). The first term is nonnegative, with equality iff X_2 is a function of X_1 (this is the basis of Rao–Blackwell).

Past problem 55 *[2019 #3]*

See Part I, cluster sampling problem (b). Stratified: every stratum is sampled; precision depends on within-stratum variance, so strata should be homogeneous internally and different from each other; gives domain estimates. Cluster: only some groups are sampled; precision depends on between-cluster variance, so clusters should be heterogeneous internally (each like the population); chosen for cost or lack of an element frame; usually less precise per element (deff > 1).

Past problem 56 *[2019 #4]*

(a) $8.4 \pm t_{0.025}(8)(2.34) = 8.4 \pm 2.306(2.34) = (3.00, 13.80)$.

(b) Error df 8, $MSE = 5.6$. $F = t^2 = (8.4/2.34)^2 = 12.89$; $SSR = F \cdot MSE = 72.16$; $SST = 116.96$; $R^2 = 0.617$.

(c) $y_i - 3.5 = \beta_1 x_i + \epsilon_i$ (regression through the origin on the shifted response): $\tilde\beta_1 = \sum x_i(y_i - 3.5)/\sum x_i^2$ with $\operatorname{Var} = \sigma^2/\sum x_i^2 \le \sigma^2/S_{xx}$, and the error has $n-1 = 9$ df. Using correct prior information increases precision; if the information is wrong the estimator is biased, and the usual R^2 is not comparable.

Past problem 57 *[2019 #5]*

(a) $EX = \phi_0/(1-\phi_1)$ and $\rho_1 = \phi_1$: $\hat\phi_1 = r_1 = \frac{\sum_{t=2}^n (x_t-\bar x)(x_{t-1}-\bar x)}{\sum_{t=1}^n (x_t-\bar x)^2}$, $\hat\phi_0 = \bar x(1-\hat\phi_1)$.

(b) The conditional likelihood $\prod_{t=2}^n \phi(x_t; \phi_0 + \phi_1 x_{t-1}, \sigma^2)$ is maximized by OLS of x_t on $(1, x_{t-1})$: $\hat\phi_1 = \frac{\sum_{t=2}^n (x_t - \bar x_{(0)})(x_{t-1}-\bar x_{(1)})}{\sum_{t=2}^n (x_{t-1}-\bar x_{(1)})^2}$, $\hat\phi_0 = \bar x_{(0)} - \hat\phi_1 \bar x_{(1)}$, where $\bar x_{(0)}, \bar x_{(1)}$ are the means of $x_2, \dots, x_n$ and $x_1, \dots, x_{n-1}$; $\hat\sigma^2 = \frac{1}{n-1}\sum \hat\epsilon_t^2$.

Past problem 58 *[2019 #6]*

(a) $X \overset{d}{=} (U_1+U_2)/2$ with $U_i \overset{\text{iid}}{\sim} U(0,1)$ (triangular density), so $M(t) = \left(\frac{e^{t/2}-1}{t/2}\right)^2 = \frac{4(e^{t/2}-1)^2}{t^2}$ for $t \ne 0$, $M(0) = 1$.

(b) $EX = \frac12$ by symmetry. Expanding $e^{t/2} - 1 = \frac{t}{2}(1 + \frac{t}{4} + \dots)$ gives $M(t) = 1 + \frac{t}{2} + O(t^2)$, so $M'(t) \to \frac12$. $M(t)$ is finite for all t (bounded support), so the mgf exists and is smooth at 0 (the apparent singularity at $t = 0$ is removable).

Past problem 59 *[2019 #7]*

(a) $P(Y > y) = P(X > \log(y/\beta)) = (\beta/y)^\alpha$ for $y \geq \beta$: Pareto, $f(y) = \alpha\beta^\alpha y^{-\alpha-1}$, $y \geq \beta$.

(b) $EY^k = \beta^k Ee^{kX} = \frac{\alpha\beta^k}{\alpha-k}$ for $k < \alpha$ (infinite otherwise). $EY = \frac{\alpha\beta}{\alpha-1}$ ($\alpha > 1$), $\operatorname{Var} Y = \frac{\alpha\beta^2}{(\alpha-1)^2(\alpha-2)}$ ($\alpha > 2$).

(c) $L = \alpha^n \beta^{n\alpha} \prod y_i^{-\alpha-1} \mathbf{1}\{\beta \leq y_{(1)}\}$ increases in β: $\hat{\beta} = Y_{(1)}$. Then $\hat{\alpha} = n/\sum \log(Y_i/Y_{(1)})$.

(d) $\hat{\beta}$ converges at rate n ($Y_{(1)}$ is Pareto with index $n\alpha$, $\operatorname{Var} Y_{(1)} \approx \beta^2/(n\alpha)^2$), so it does not affect the first-order behaviour of $\hat{\alpha}$: $\operatorname{Var}\hat{\alpha} \approx \alpha^2/n$ ($I(\alpha) = 1/\alpha^2$). (Exactly, $2n\alpha/\hat{\alpha} \sim \chi^2_{2(n-1)}$.)

Past problem 60 *[2019 #8]*

Statement. For $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$, a test ϕ^* with $\phi^* = 1$ if $f_1(\mathbf{x}) > kf_0(\mathbf{x})$, $\phi^* = 0$ if $f_1(\mathbf{x}) < kf_0(\mathbf{x})$ and $E_0\phi^* = \alpha$ is most powerful among tests with $E_0\phi \leq \alpha$; such a test exists for every α (with randomization on $f_1 = kf_0$).

Proof. For any ϕ with $0 \leq \phi \leq 1$ and $E_0\phi \leq \alpha$, $(\phi^* - \phi)(f_1 - kf_0) \geq 0$ pointwise (where $f_1 > kf_0$, $\phi^* = 1 \geq \phi$; where $f_1 < kf_0$, $\phi^* = 0 \leq \phi$). Integrating, $E_1\phi^* - E_1\phi \geq k(E_0\phi^* - E_0\phi) \geq 0$.

Past problem 61 *[2019 #9 (essay)]*

Outline. (1) Problems: inconsistent concepts, definitions, classifications and coverage (double counting or gaps); different reference periods, valuation (basic vs market prices), seasonal-adjustment methods and revision schedules; different quality and sampling errors that propagate and are hard to quantify; aggregates that do not equal the sum of parts (chain-linking non-additivity, separate seasonal adjustment of components vs direct adjustment of the total). (2) Causes: each source designed for its own purpose; lack of coordination and metadata. (3) Actions: reconcile definitions with a common framework (SNA, standard classifications), bridge tables for coverage, benchmarking and balancing (e.g. Denton, RAS) to enforce accounting identities, consistent seasonal adjustment policy (direct vs indirect), document sources and quality, coordinate release and revision calendars with providers, evaluate the error contribution of each source.